Deep Networks for Direction-of-Arrival Estimation in Low SNR

Georgios K. Papageorgiou, *Member, IEEE*, Mathini Sellathurai, *Senior Member, IEEE*, and Yonina C. Eldar, *Fellow, IEEE*

***Abstract*—In this work, we consider direction-of-arrival (DoA) estimation in the presence of extreme noise using Deep Learning (DL). In particular, we introduce a Convolutional Neural Network (CNN) that predicts angular directions using the sample covariance matrix estimate. The network is trained from multi-channel data of the true array manifold matrix in the low signal-to-noise-ratio (SNR) regime. By adopting an on-grid approach, we model the problem as a multi-label classification task and train the CNN to predict DoAs across all SNRs. The proposed architecture demonstrates enhanced robustness in the presence of noise, and resilience to a relatively small number of snapshots. Moreover, it is able to resolve angles within the grid resolution. Experimental results demonstrate significant performance gains in the low-SNR regime compared to state-of-the-art methods and without the requirement of any parameter tuning in both cases of correlated and uncorrelated sources. Finally, we relax the assumption that the number of sources is known *a priori* and present a training method, where the CNN learns to infer their number and predict the DoAs with high confidence. The increased robustness of the proposed solution is highly desirable in challenging scenarios that arise in several fields, ranging from wireless array sensors to acoustic microphones or sonars.**

***Index Terms*—Direction-of-arrival (DoA) estimation, convolution neural network CNN, deep learning DL, multilabel classification, array signal processing.**

I. Introduction

DIRECTION-OF-ARRIVAL (DoA) estimation has been at the forefront of research activity for many decades, due to the plethora of applications ranging from radar and wireless communications to sonar and acoustics [1], with localization being one of the most significant ones. Estimation of the angular directions is possible with the use of multiple sensors in a specified geometric configuration, e.g., linear, rectangular and circular. Efficient use of the observations from multiple sensors enables the DoA estimation of several sources, depending on the number of array sensors. There are two major categories in DoA estimation: the overdetermined case, where the number of sources is less than the number of array sensors, and the underdetermined case, where the number of sources is equal or greater than the number of sensors [2], [3]. In this work, we investigate the first category. Moreover, we focus on DoA estimation in low SNR, which poses several challenges and is critically important in many real-world situations.

One of the first methods introduced for DoA estimation is MUltiple SIgnal Classification (MUSIC) [4], with several other variants following soon after. The MUSIC estimator belongs to the class of subspace-based techniques, which attempts to separate the signal and noise sub-spaces; angle estimation follows from the so-called MUSIC pseudo-spectra over a specified grid, where the corresponding peaks of the pseudo-spectra are selected. Estimation of signal parameters via rotational invariance techniques (ESPRIT) [5] and its variants is another successful example [6]. Unitary ESPRIT [6], [7], which incorporates forward-backward averaging, is a notable variant that leads to improved performance compared to ESPRIT, especially in cases of correlated source signals. A significant step towards improvement in DoA estimation was with the development of Root-MUltiple SIgnal Classification (R-MUSIC), which estimates the angular directions from the solutions of higher-order polynomials [8]. The aforementioned methods, are covariance-based techniques that require a sufficient number of data snapshots to accurately estimate the DoAs, particularly in low SNRs. Furthermore, they often assume that the number of sources is known, which is not the case in many practical applications.

During the past decade, Compressed Sensing (CS) methodologies have also been used to address DoA estimation [9], [10]. These methods exploit the sparse characteristic of the signal sources in the spatial domain (angles). CS techniques are generally separated into three main categories: a) on-grid, b) off-grid and c) grid-less methods [11], [12]. Grid-less approaches achieve better performance at the expense of very high computational complexity [13]. Off-grid and on-grid methods offer a more balanced solution with less computational demands at the expense of negligible loss in performance due to the grid mismatch problem [14]. DoA estimates are obtained after the solution of sparse minimization tasks, for which two major approaches are identified: i) greedy methods based on the ℓ_0 pseudo-norm and ii) convex relaxations based on the ℓ_1-norm. Notable is the method known as $\ell_{2,1}$-SVD, which first performs a dimensionality reduction technique to the received signal data

Manuscript received November 16, 2020; revised April 26, 2021; accepted June 1, 2021. Date of publication June 16, 2021; date of current version July 13, 2021. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Florian Roemer. This work was supported in part by UK's EPSRC Grant EP/P009670/1. *(Corresponding author: Georgios Papageorgiou.)*

Georgios K. Papageorgiou and Mathini Sellathurai are with the School of Engineering and Physical Sciences, Heriot-Watt University, Edinburgh, U.K. (e-mail: g.papageorgiou@hw.ac.uk; m.sellathurai@hw.ac.uk).

Yonina C. Eldar is with the Electrical Engineering, Technion-Israel Institute of Technology, Haifa 32000, Israel (e-mail: yonina.eldar@weizmann.ac.il).

This article has supplementary downloadable material available at https://doi.org/10.1109/TSP.2021.3089927, provided by the authors.

Digital Object Identifier 10.1109/TSP.2021.3089927

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

and then solves the $\ell_{2,1}$-norm minimization task in the reduced dimension with significantly less computational burden. It was introduced in [15] and was also employed later in [16]. One of the major disadvantages that all these methods have in common is that tuning of one or more parameters (which depends on the number of snapshots, the SNR or both) is required to guarantee good performance; moreover, the DoA estimates are often extremely sensitive to the tuning of these parameters.

A very recent approach to DoA estimation is via the use of Deep Learning (DL) [17], [18]. DL-based methods enjoy several advantages over optimization-based ones: a) after training the network no optimization is required and the solution is the result of simple operations (multiplications and additions); b) they do not require any specific tuning of parameters, in contrast to optimization-based techniques, whose solution strongly depends on the tuning of those parameters, and c) they demonstrate resilience to data imperfections, e.g., using fewer snapshots, performing well in low SNR. A deep neural network (DNN) with fully connected (FC) layers was employed in [19] for DoA classification of two targets using the signal covariance matrix. However, the reported results indicate poor DoA estimation results in high SNR. The authors in [20] proposed a DNN for channel estimation in massive MIMO systems; however, they focus on the high SNR regime and on channel estimation performance. A multilayer autoencoder with a series of parallel multilayer classifiers, i.e., a multi-layer perceptron (MLP), was employed in [21] with focus on the robustness to array imperfections. The MLP architecture addresses DoA estimation of only two sources via the use of a multitask autoencoder that acts as a group of spatial filters followed by a series of paralllel multi-layer DNNs for spatial spectrum estimation. The network is trained at each individual SNR.[1] A deep Convolutional Neural Network (CNN) that was also trained in low SNRs was proposed in [22]. However, the method did not demonstrate significant performance gains in terms of DoA estimation, due to the adoption of 1-dimensional (1D) filters (convolutions). A CNN for broadband DoA estimation in the context of speech processing was proposed in [23], [24]. In contrast to these works, we study the case of narrow-band DoA estimation. The authors in [25], [26] employed DNNs for the range-based localization of ships. Their approach is based on distance estimation, whereas we focus on the estimation of the signal's directions. A DL-based method for pseudo-spectra estimation was published in [27], where the authors also proposed an extension to estimate the angles. However, the demonstrated results were in high SNR and the number of sources was assumed to be known. In the context of acoustics, the authors in [28] proposed a DNN for beamforming with a single-snapshot sample covariance matrix (SCM), which was later extended in [29], [30] to include slightly more snapshots and sources. Such an approach cannot be adopted in low SNR scenarios, where the number of snapshots and sensors needs to be considerably higher.

To the best of our knowledge *no specific DL-based technique was developed for robust DoA estimation in the low SNR regime*. Another disadvantage of methods such as [21], [22] is that they are trained for a specific number of snapshots, which leads to significant deviations for different amount of snapshots and varying SNR.

The scope of this work is to fill in the gap in the literature of DoA estimation in the low SNR with the use of DL. Due to large deviations of the SCM estimates from the true manifold matrix in low SNR, DoA estimation becomes really challenging and the majority of the methods fail to demonstrate the desired robustness. In this work, we contribute towards this direction by: a) exploiting a deep network with 2D convolutional layers, which are well known for their excellent feature extraction properties; b) using multi-channel data, i.e., the real part, imaginary part, and phase of the complex valued covariance matrix entries (similar to how they are used in image processing); and c) employing dropout layers as a means of regularization, thus, improving the generalization of the estimator and avoiding over-fitting. Our contributions are summarized as follows:

- We introduce a deep CNN trained on multi-channel data, which are explicitly formed from the complex-valued data of the *true covariance matrix*. The first and second channels are formed by taking the real and imaginary parts, respectively, from the complex-valued entries of the covariance matrix. The third channel includes the phase from the complex valued entries of the covariance matrix (in $[-\pi, \pi]$). The proposed CNN employs 2D convolutional layers and is trained to directly predict the angular directions of multiple sources. Testing is performed using the SCM estimate for any amount of snapshots. The use of multi-channel data along with the adoption of 2D convolutional layers enables the extraction of features from the input data leading to more robust DoA estimation in low SNR. A discretization approach (on-grid) is adopted for the desirable angular (spatial) region and the direction estimation task is modeled as a multi-label classification one.
- Efficient methods for training the proposed network are presented. In particular, we train the CNN across a range of low SNRs and demonstrate that it can successfully predict DoAs in high SNR as well.
- We introduce a training method for a varying number of sources. Subsequently, *the proposed CNN infers the number of sources from the received data, while predicting the DoAs*.
- The performance of the proposed solution is evaluated over an extensive set of simulated experiments, where it is compared against state-of-the-art methods in various experimental set-ups with off-grid angles. Additionally, comparison to the Cramér-Rao lower bound (CRLB) is provided as benchmark.

The results indicate that the proposed CNN: a) outperforms its competitors in DoA estimation in the low SNR regime; b) is resilient in estimation even for a small collection of snapshots regardless of the angular separation of the sources; c) demonstrates enhanced robustness in case of SNR mismatches, and d) is able to infer both the number of sources and DoAs with very small errors and high confidence level.

The rest of the paper is organized as follows: in Section II, we present the signal model. In Section III, we introduce the

[1]https://github.com/LiuzmNUDT/DNN-DOA

proposed CNN for DoA prediction and in Section IV we discuss the adopted training approach. Section V presents simulation results and in Section VI, we summarize and highlight our conclusions.

Notation: Throughout the paper the following notation is adopted: $\mathcal{X}$ denotes a set and $|\mathcal{X}|$ its cardinality; $\mathbf{X}$ is a matrix, $\mathbf{x}$ is a vector and x is a scalar. The i,j-th element of a matrix $\mathbf{X}$ is denoted $(\mathbf{X})_{i,j}$ and the i-th entry of a vector $\mathbf{x}$ is $x(i)$. The i-th example of the vector $\mathbf{x}$ is denoted as $\mathbf{x}(i)$. The imaginary unit is j (so that $\mathrm{j}^2 = -1$). The conjugate transpose of a matrix is $(\cdot)^H$; its conjugate is $(\cdot)^*$ and its transpose is $(\cdot)^T$. The $N \times N$ identity matrix is $\mathbf{I}_N$. The white circularly-symmetric Gaussian distribution with mean $\mathbf{m}$ and covariance $\mathbf{C}$ is denoted by $\mathcal{CN}(\mathbf{m}, \mathbf{C})$. The convolution operator is denoted as $*$. The floor of a number α is written as $\lfloor \alpha \rfloor$. Functions are denoted by lower case italics, e.g., $f(\cdot)$. The symbol $\mathbb{E}[\cdot]$ is the expectation operator; $\mathrm{Re}\{\cdot\}, \mathrm{Im}\{\cdot\}$ denote the real and the imaginary parts of a complex scalar / vector / matrix, respectively. Finally, the phase of the complex-valued variable α is denoted by $\angle\{\alpha\}$.

II. Signal Model

The standard model for a N-element sensor array in the narrow-band mode, with K far-field sources present, is:

$$\mathbf{y}(t) = \sum_{k=1}^{K} \mathbf{a}(\theta_k) s_k(t) + \mathbf{e}(t) = \\ = \mathbf{A}(\boldsymbol{\theta})\mathbf{s}(t) + \mathbf{e}(t),\ t = 1, \ldots, T. \quad (1)$$

Here $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \ldots, \mathbf{a}(\theta_K)]$ is the $N \times K$ array manifold matrix, $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_K]^T$ is the vector of (unknown) source directions and T is the total number of collected snapshots, $\mathbf{s}(t) = [s_1(t), \ldots, s_K(t)]^T$ and $\mathbf{e}(t)$ denote the transmitted signal and additive noise vectors at sample index t, respectively. The model (1) is generic in the sense that it does not depend on the array geometry; however, in this work we will consider a uniform linear array (ULA) configuration for simplicity.[2] Thus, the columns of the array manifold matrix are expressed as

$$\mathbf{a}(\theta_k) = [1, e^{\mathrm{j}\frac{2\pi d}{\lambda}\sin(\theta_k)}, \ldots, e^{\mathrm{j}\frac{2\pi d}{\lambda}\sin(\theta_k)(N-1)}]^T, \quad (2)$$

where d is the array interelement distance and $\lambda = c/f$ is the wavelength at carrier frequency f with c the speed of light/sound. In this case, $\mathbf{A}(\boldsymbol{\theta})$ becomes a Vandermonde matrix. The following assumptions are typical in the literature of narrow-band DoA estimation:

A1) The source DoAs are distinct.

A2) Each source signal follows the unconditional-model assumption (UMA) in [31], which assumes that the transmitted signal is randomly generated (Gaussian signaling). Moreover, the sources are uncorrelated, leading to a diagonal source covariance matrix: $\mathbf{R}_s = \mathbb{E}[\mathbf{s}(t)\mathbf{s}^H(t)] = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_K^2)$.

A3) The additive noise values are independent and identically distributed (i.i.d.) zero-mean white circularly-symmetric Gaussian, i.e., $\mathbf{e}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I}_N)$ and uncorrelated from the sources.

A4) There is no temporal correlation between each snapshot.

We are interested in the estimation of the unknown DoAs $\boldsymbol{\theta}$ from measurements $\mathbf{y}(1), \ldots, \mathbf{y}(T)$. Considering A1–A4, the received signal's covariance matrix is given by:

$$\mathbf{R}_y = \mathbb{E}[\mathbf{y}(t)\mathbf{y}^H(t)] = \mathbf{A}(\boldsymbol{\theta})\mathbf{R}_s\mathbf{A}^H(\boldsymbol{\theta}) + \sigma_e^2\mathbf{I}_N. \quad (3)$$

The statistical richness of $\mathbf{R}_y$ in (3) allows for the estimation of up to $K \leq N-1$ distinct DoAs. However, in practice, the matrix in (3) is unknown and is replaced by its sample estimate

$$\widetilde{\mathbf{R}}_y = \frac{1}{T}\sum_{t=1}^{T} \mathbf{y}(t)\mathbf{y}^H(t), \quad (4)$$

which is an unbiased estimator of $\mathbf{R}_y$.

We note that assumptions A2–A4 are only used for generating the data in the training procedure of the proposed CNN. The network can make predictions even if these assumptions are violated. However, as the deviation of the test from the training data becomes higher, estimation becomes less accurate. Violation of these assumptions has different impact on classical estimators in the field of DoA estimation as well.

III. A Deep Convolutional Neural Network for DoA Estimation

In this section, we formulate DoA estimation as a multilabel classification task. In Section III-A, we present the data management and labeling approach, whereas Section III-B is devoted to the description of the CNN's architecture that learns to predict the DoAs. The convolution layers perform the feature extraction from the multi-channel input data, and, subsequently, the FC layers use the output of the convolution layers to infer the DoA estimates using a pre-selected grid.

A. Data Management and Labeling

DoA prediction is modeled as a *multilabel classification task*. For $\phi_{\max} \in \{1^\circ, \ldots, 90^\circ\}$ we consider $2G+1$ discrete points of resolution ρ (in degrees), which define a grid $\mathcal{G} = \{-G\rho, \ldots, -\rho, 0^\circ, \rho, \ldots, G\rho\} \subset [-90^\circ, 90^\circ]$, such that $\phi_{\max} = G\rho$. At each SNR level, K angles are selected from the set $\mathcal{G}$ and the respective covariance matrix is calculated according to (3). The input data $\mathbf{X}$ of the proposed CNN is a real-valued $N \times N \times 3$ matrix, whose third dimension represents different "channels." In particular, the first and second channels are the real and imaginary parts of $\mathbf{R}_y$, i.e., $\mathbf{X}_{:,:,1} = \mathrm{Re}\{\mathbf{R}_y\}$ and $\mathbf{X}_{:,:,2} = \mathrm{Im}\{\mathbf{R}_y\}$, whereas the third channel corresponds to phase entries, i.e., $\mathbf{X}_{:,:,3} = \angle\{\mathbf{R}_y\}$. Thus, the input to the CNN is a collection of D data points defined as $\mathcal{X} = \{\mathbf{X}_{(1)}, \ldots, \mathbf{X}_{(D)}\}$.

Next, for each example $\mathbf{X}_{(i)}$, the K training angles in $\mathcal{G}$ are transformed into a binary vector with K ones (the rest are zeros). For example, if $\phi_{\max} = 60^\circ$ and the desired resolution is $\rho = 1^\circ$, the grid becomes $\mathcal{G} =$

[2]The analysis and methodology also holds for any other array configuration, e.g., non-uniform linear or rectangular.

$\{-60^\circ, \ldots, -1^\circ, 0^\circ, 1^\circ, \ldots, 60^\circ\}$ with $|\mathcal{G}| = 121$ grid points; moreover, the angle pair $\{-60^\circ, -59^\circ\}$ corresponds to the 121×1 binary vector $\mathbf{z} = [1, 1, 0, \ldots, 0]^T$, which serves as the corresponding label/output of the proposed CNN. Thus, the i-th label $\mathbf{z}_{(i)}$ belongs to the set $\mathcal{Z} = \{0, 1\}^{2G+1}$ according to the described process. Hence, the i-th training example consists of pairs in the form $(\mathbf{X}_{(i)}, \mathbf{z}_{(i)})$ leading to the training data set $\mathcal{D} = \{(\mathbf{X}_{(1)}, \mathbf{z}_{(1)}), (\mathbf{X}_{(2)}, \mathbf{z}_{(2)}), \ldots, (\mathbf{X}_{(D)}, \mathbf{z}_{(D)})\}$ of size D.

According to the well-known universal approximation theorem [32] a feed-forward network with a single hidden layer processed by a multilayer perceptron can approximate continuous functions on a compact subset of $\mathbb{R}^n$. The goal of this multilabel classification task is to induce a ML hypothesis defined as a function f from the input space to the output space, i.e., $f : \mathbb{R}^{N \times N \times 3} \to \mathcal{Z}$. Although the true covariance matrix is used for training the network, for its testing and evaluation the sample covariance in (4) is used, since the former is unknown. To this end, during the testing phase of the CNN all input examples can be considered as "unseen data" to the training.

B. The Proposed CNN's Architecture

The nonlinear function f is parametrized by a CNN of 8 layers, i.e.,

$$f(\mathbf{X}) = f_8(f_7(\ldots f_1(\mathbf{X}))) = \mathbf{z}. \tag{5}$$

The architecture of the proposed CNN is based on the standard convolutional structures used in the literature of image processing [33], [34] with some modifications that are required, due to the nature of our problem. Each function $\{f_i(\cdot)\}_{i=1,\ldots,4}$ represents a series of convolutional layers: a 2D convolutional layer of $n_C = 256$ filters, followed by a batch normalization layer [35] and a rectified linear unit `ReLU` layer, i.e., the nonlinear activation function $\texttt{ReLU}(x) = \max(0, x)$ applied element wise to the variables of the previous layer. Additionally, after the `ReLU` layer of $f_4(\cdot)$ a flatten layer is used, which shapes the tensor-valued output of the final convolutional layer to a vector. For the kernel of size $\kappa \times \kappa$, we used $\kappa = 3$ for $f_1(\cdot)$ and $\kappa = 2$ for the rest of the convolutional layers. The stride δ that we used is $\delta = 2$ for $f_1(\cdot)$ and $\delta = 1$ for the rest of the convolutional layers (no padding). Hence, for each one of the n_C filters the mathematical expression of the convolution operation at the first layer with input data $\mathbf{X} \in \mathbb{R}^{N \times N \times 3}$ and kernel $\mathbf{K} \in \mathbb{R}^{\kappa \times \kappa \times 3}$ is a 2D matrix of dimension $M \times M$ (output of the layer per filter) given by:

$$(\mathbf{X} * \mathbf{K})_{m,n} = \sum_{i=1}^{\kappa} \sum_{j=1}^{\kappa} \sum_{k=1}^{3} K_{i,j,k} X_{\delta(m-1)+i, \delta(n-1)+j, k}, \tag{6}$$

$m, n = 1, \ldots, M$, where $M = \lfloor (N - \kappa)/\delta \rfloor + 1$. Thus, the convolution operation of the q-th filter at the ℓ-th convolutional layer has the following parameters:

- **Input**: $\mathbf{X}^{[\ell-1]}$ of size $M^{[\ell-1]} \times M^{[\ell-1]} \times n_C^{[\ell-1]}$ with $\mathbf{X}^{[0]} = \mathbf{X}$ and $M^{[0]} = N$, $n_C^{[0]} = 3$;
- **Filter**: kernel $\mathbf{K}_q^{[\ell]}$ of dimension $\kappa^{[\ell]} \times \kappa^{[\ell]} \times n_C^{[\ell-1]}$;
- **Stride**: $\delta^{[\ell]}$;
- **Bias**: $b_q^{[\ell]}$;
- **Output**: $\mathbf{X}_q^{[\ell]} = \mathbf{X}^{[\ell-1]} * \mathbf{K}_q^{[\ell]}$ of size $M^{[\ell]} \times M^{[\ell]}$, and is the $M^{[\ell]} \times M^{[\ell]}$ matrix given by:

$$(\mathbf{X}^{[\ell-1]} * \mathbf{K}_q^{[\ell]})_{m,n} = \sum_{i=1}^{\kappa^{[\ell]}} \sum_{j=1}^{\kappa^{[\ell]}} \sum_{k=1}^{n_C^{[\ell-1]}} K_{i,j,k}^{q,[\ell]} X_{\delta(m-1)+i, \delta(n-1)+j, k}^{[\ell-1]} + b_q^{[\ell]}, \tag{7}$$

for $m, n = 1, \ldots, M$ and $q = 1, 2, \ldots, n_C^{[\ell]}$. The collection of the outputs in (7) for all filters $q = 1, 2, \ldots, n_C^{[\ell]}$ leads to the tensor $\mathbf{X}^{[\ell]}$ of dimension $M^{[\ell]} \times M^{[\ell]} \times n_C^{[\ell]}$. Thus, the total number of learned parameters at the ℓ-th layer are $(\kappa^{[\ell]} \times \kappa^{[\ell]} \times n_C^{[\ell-1]}) \times n_C^{[\ell]}$ for the filters plus $n_C^{[\ell]}$ for the biases. Pooling layers were not used (although tested), since the loss of information resulted in poor performance.

Thereafter, the FC layers follow. Each function $\{f_i(\cdot)\}_{i=5,6,7}$ is a dense layer with 4096, 2048 and 1024 neurons, respectively, followed by a `ReLU` layer and a Dropout layer. The latter randomly sets weights to zero with probability 30% (non trainable parameters), so the network is forced to learn instead of memorizing the data. The Dropout layers also act as regularization to the learning process. The ℓ-th FC layer maps its input $\mathbf{c}^{[\ell-1]} \in \mathbb{R}^{V^{[\ell-1]}}$ to the output $\mathbf{c}^{[\ell]} \in \mathbb{R}^{V^{[\ell]}}$ via a set of weights $\mathbf{W}^{[\ell]} \in \mathbb{R}^{V^{[\ell]} \times V^{[\ell-1]}}$ and biases $\mathbf{b}_{\text{FC}}^{[\ell]} \in \mathbb{R}^{V^{[\ell]}}$. Thus, the output of the ℓ-th FC layer (before nonlinear activation and dropout) is given by

$$\mathbf{c}^{[\ell]} = \mathbf{W}^{[\ell]} \mathbf{c}^{[\ell-1]} + \mathbf{b}_{\text{FC}}^{[\ell]}, \tag{8}$$

where the set of parameters $\vartheta^{[\ell]} = \{\mathbf{W}^{[\ell]}, \mathbf{b}_{\text{FC}}^{[\ell]}\}$ with $V^{[\ell-1]} \times V^{[\ell]} + V^{[\ell]}$ entries (in total) is optimized during the training of the neural network. The final (output) layer, $f_8(\cdot)$, consists of a dense layer with $2G + 1$ neurons followed by a `Sigmoid` layer, which applies the function $s(x) = e^x/(e^x + 1)$ element-wise to the values of the previous layer and returns values in [0, 1]. The selection of the sigmoid function over the softmax is due to the presence of K labels, which could independently receive a value equal (during the training) or close (during the inference) to 1.

Thus, the output of the CNN is a probability at each entry of the predicted label, which for the input data $\mathbf{X}_{(i)}$ is expressed as

$$\hat{\mathbf{p}}_{(i)} = f(\mathbf{X}_{(i)}) = \begin{pmatrix} \hat{p}_1 \\ \vdots \\ \hat{p}_{2G+1} \end{pmatrix}. \tag{9}$$

The layout of the proposed CNN is depicted in Fig. 1.

The training of the CNN is performed offline in a supervised manner over the training data set $\mathcal{D}$. In particular, since the adopted approach is a multilabel classification task, we attempt to optimize the set of all trainable parameters ϑ whose updates are carried out via back-propagation by minimizing the reconstruction error, i.e.:

$$\vartheta^* = \arg\min_{\vartheta} \frac{1}{D} \sum_{i=1}^{D} \mathrm{L}\left(\hat{\mathbf{p}}_{(i)}; \mathbf{z}_{(i)}\right), \tag{10}$$

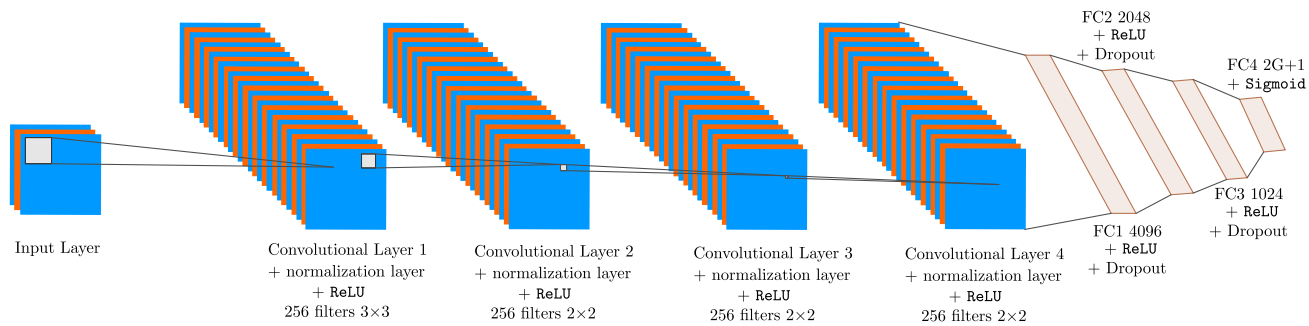


Fig. 1. The architecture of the proposed CNN for DoA prediction in low SNR. Each convolutional layer has 256 filters and is followed by a batch normalization layer and a ReLU layer. After the convolutional layers a flatten layer is used that transfers the learning process to fully connected (FC) layers (followed by ReLU activation functions), which lead to the classification probability induced by the Sigmoid activation function at the final layer. Dropout layers are also employed (for regularization) after each of the first three FC layers.

where

$$\mathrm{L}(\hat{\mathbf{p}}_{(i)};\mathbf{z}_{(i)}) = -\frac{1}{2G+1}\sum_{n=1}^{2G+1}[z_{(i)}(n)\log(\hat{p}_{(i)}(n)) + (1-z_{(i)}(n))\log(1-\hat{p}_{(i)}(n))], \quad (11)$$

is the binary cross-entropy loss.

IV. Training Approach

For training of the proposed CNN we considered a grid of $\rho = 1^\circ$ resolution and considered two cases: i) using $\phi_{\max} = 60^\circ$ (or $G = 60$) which defines the narrow grid set $\mathcal{G}_n = \{-60^\circ, \ldots, 60^\circ\}$ with $|\mathcal{G}_n| = 121$ grid points, and ii) using $\phi_{\max} = 90^\circ$ (or $G = 90$) which defines the wide grid set $\mathcal{G}_w = \{-90^\circ, \ldots, 90^\circ\}$ with $|\mathcal{G}_w| = 181$ grid points. Hence, the output dimension of the CNN after the binary transformation of the DoAs is either 121 or 181 (depending on $\phi_{\max}$). In each case, the proposed network was trained separately. We resort to data from the true covariance matrix in order to: a) reduce the number of training data required and b) enable the DoA prediction for various snapshots of the SCM estimate. One of the main advantages of the proposed approach is that the CNN can also be trained to infer the number of sources, since the problem is modeled as a multi-classification task. Therefore, we adopt two approaches for the training of the CNN: in Section IV-A we consider *a fixed number of sources*, which is more suitable in cases where the number of sources is known a priori. In Section IV-B, the network is trained for a varying number of sources; therefore, we will refer to the second one as a *mixed number of sources* training approach, which is more suitable in cases where *the number of sources/targets is unknown*, such as in military applications. The experimental section, where the proposed model is evaluated, is also separated into two parts, according to the training options described next. In both approaches training is performed offline from synthetic data generated using on-grid angles (on either $\mathcal{G}_n$ or $\mathcal{G}_w$) and the true covariance matrix in (3).

A. Fixed Number of Sources

In this section, we describe the synthetic data generation process for the case where the number of sources is known *a priori*. We consider $K = 2$ and generate the pairs of DoAs from all possible combinations in $\mathcal{G}$, i.e., $\binom{2G+1}{K}$ pairs of angles.[3] After generating the $\binom{2G+1}{K}$ pairs of angles at each SNR, we first calculate the corresponding input data $\mathbf{X}$ using the true covariance matrix in (3) and the labels as described in Section III-A. For the narrow angular region we have $G = 60$, leading to 7260 training examples generated at each SNR, whereas for $G = 90$ we have 16290 training examples per SNR value. For the training of the proposed CNN we used data from low SNRs, i.e., $\{-20, -15, -10, -5, 0\}$ dB leading to a total of $D = 5 \cdot 7260 = 36300$ and $D = 5 \cdot 16290 = 81450$ training examples in total for the narrow and wide range, respectively. We observed that training in the low SNR regime (worst case scenario training approach) is also sufficient for prediction at higher SNRs. Once the CNN is trained prediction can be performed at high SNRs, as we experimentally demonstrate in Section V-B. Although training at each individual SNR could slightly improve the performance of the proposed CNN (we tested this but the improvement is not significant), it can be quite limiting: a) several models per SNR need to be trained and their parameters stored and b) in the testing phase the operating SNR should be known (which is typically not the case). Therefore, we resorted to the joint training in the range of low SNRs.

The training of the CNN is performed offline. The data were randomly split into training (90%) and validation sets (10%). In Fig. 2(a), (b) we have plotted the training and validation binary accuracy (top) and loss (bottom) at each epoch for the fixed number of sources, where (a) corresponds to the narrow grid and (b) corresponds to the wide one. Since the validation loss is lower than the training loss no over-fitting occurs. However, it should be noted that the validation set consists of on-grid angles only. However, this is not the case for the testing data, which are produced a) from off-grid angles and b) using the SCM estimate (for a selected number of snapshots) instead of the true covariance matrix which is in practice unknown. For the update/optimization of the CNN's parameters we employed Adam [36] with an initial learning rate of 0.001, whereas $\beta_1 = 0.9$, $\beta_2 = 0.999$; additionally, the learning rate was scaled by a

[3]The order of the DoAs does not play any role, since the covariance matrix (input to the CNN) remains the same for the given angles.

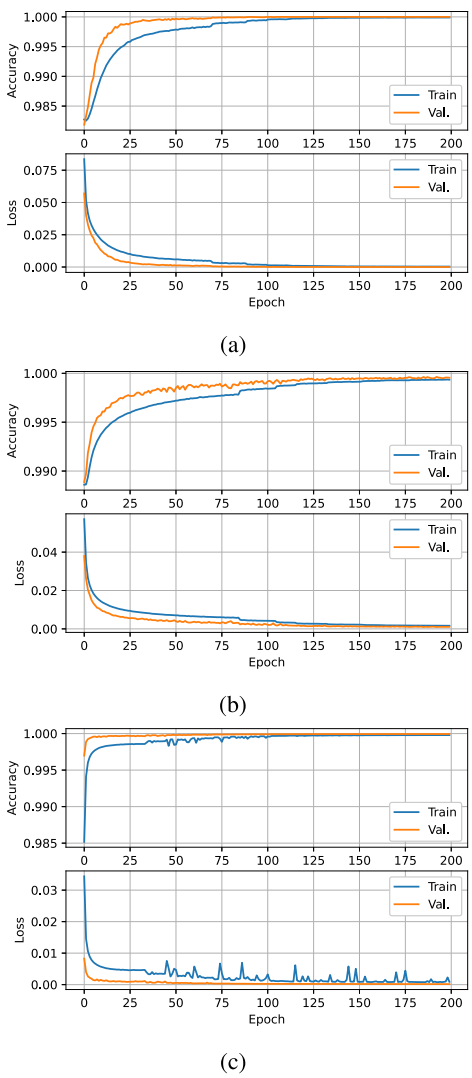


Fig. 2. The training and validation binary accuracy (top) and loss (bottom) of the CNN for each training approach: (a) fixed number of sources and narrow grid $\mathcal{G}_n$, (b) fixed number of sources and wide grid $\mathcal{G}_w$, and (c) mixed/varying number of sources and narrow grid $\mathcal{G}_n$.

factor of 0.7 (reduced) once the validation loss reached plateau with a patience of 10 epochs. The batch size was set to 32 and the network was trained for 200 epochs. The CNN was implemented in Keras using Tensorflow as backend; the operating system was Windows running on an Intel Xeon Gold 5222 processor at 3.8 GHz and with an NVIDIA TITAN RTX GPU.

B. Mixed Number of Sources

Here, we describe our training approach for the mixed number of sources, which varies from 1 to $K_{\max}$ and $G = 60$ (narrow grid). In the experiments (Section V-C) we have used $K_{\max} = 3$. The training data at each SNR and for each $k = 1, \ldots, K_{\max}$ are generated as described in Section IV-A. The difference now is that instead of having 7260 training examples at each SNR, we have $\sum_{k=1}^{K_{\max}} \binom{2G+1}{k} = 295361$ training examples (summation for all $k = 1, \ldots, K_{\max}$). Each training label is an 121×1 binary vector with the number of 1's varying from 1 to $K_{\max} = 3$ (rest of entries are zero). Training was performed offline in a similar manner with data from low SNRs $\{-15, -10, -5, 0\}$ dB[4] using the true covariance matrix. Thus, a total of $D = 4 \cdot 295361 = 1181444$ training examples were used. The fact that K now varies makes the learning process more demanding, as is depicted in Fig. 2(c). The network's training parameters were the same as before, except for the batch size, which was increased to 64.

Although standard methods in the literature of DoA estimation (e.g., MUSIC,) can estimate up to $K = N - 1$ sources/targets, this is not straightforward with the use of neural networks. The size of the training data increases exponentially (does not scale). We should note that this is not a limitation of the proposed CNN only but rather a challenge that most DL-based approaches face, due to the need to perform exhaustive training [19], [21], [22]. In addition, under the particular scenario of low SNR, estimation of $N - 1$ is not very accurate even for standard methods such as MUSIC, EPSRIT and R-MUSIC.

V. Simulation Results

In this section, we provide extensive simulation results, where we evaluate the performance of the proposed CNN in the DoA estimation under various setups. First, we provide a summary of the methods that the proposed CNN was compared to, where we evaluate the performance of the CNN assuming that the number of sources is known. In the second part, after training the network for a mixed number of sources (up to a maximum number), we relax the assumption that the number of sources is known and provide results where the CNN is able to predict both the number of sources and the DoAs.

In both training and testing experiments phases, we consider a ULA with $N = 16$ elements equally spaced at half-wavelength distance ($d = \lambda/2$). The number of trainable parameters for the first convolutional layer are 7680, whereas each one of the second, third and fourth convolutional layers has 262912 parameters (including 512 trainable parameters corresponding to each of the four batch normalization layers). The first FC layer has 16781312 trainable parameters; the second one has 8390656; the third one has 2098176 and the fourth one has 124025 for $G = 60$ and 185525 for $G = 90$ (depending on the selection of narrow or wide grid). In total, the proposed CNN has approximately 28.2 million parameters that need to be trained. For all the experiments, the SNR is defined as in [37]:

$$\mathrm{SNR} = 10 \log_{10} \frac{\min\{\sigma_1^2, \sigma_2^2, \ldots, \sigma_K^2\}}{\sigma_e^2}. \tag{12}$$

After training, the CNN predicts

$$\hat{\mathbf{p}} = f(\widetilde{\mathbf{X}}) = \begin{pmatrix} \hat{p}_1 \\ \vdots \\ \hat{p}_{2G+1} \end{pmatrix}, \tag{13}$$

[4]Data from -20 dB was excluded since it is difficult to make highly accurate DoA predictions there even if K is known; as we have seen in practice, this holds for all estimators and not only the proposed CNN.

where $\widetilde{\mathbf{X}} \in \mathbb{R}^{N\times N\times 3}$ with $\mathbf{X}_{:,:,1} = \mathrm{Re}\{\widetilde{\mathbf{R}}_y\}$, $\mathbf{X}_{:,:,2} = \mathrm{Im}\{\widetilde{\mathbf{R}}_y\}$ and $\mathbf{X}_{:,:,3} = \angle\{\widetilde{\mathbf{R}}_y\}$.

A. Comparison to Other Methods

In the following, we summarize the methods that we used in the comparison to the proposed CNN:

1) MUSIC in [4].
2) R-MUSIC in [8].
3) ESPRIT in [5], [38].
4) Unitary ESPRIT in [6], [7], [38] denoted as UnESPRIT.
5) $\ell_{2,1}$-SVD in [15], [16].
6) MLP in [21].

The methods 1-4) correspond to classical methods in the literature of DoA estimation; method 5) is a good representative of CS-based methods, whereas 6) is used in the comparison to recent DL-based approaches (also see Section I for further details). For the on-grid methods (MUSIC and $\ell_{2,1}$-SVD) the adopted grid resolution is chosen to be the same as that of the CNN ($\rho = 1^\circ$). For both ESPRIT and UnESPRIT the maximum subarray overlap was used as well as the row weighting technique with the maximum possible weight, which in our case equals $N/2 = 8$; furthermore, in both implementations the total least-squares (TLS) criterion was used [38]. Additionally, we have calculated two CRLBs: CRLB_{uncr}, which denotes the bound for strictly uncorrelated sources introduced in [39] and CRLB that is based on the unconditional model introduced in [31] for the case of correlated sources. We should note that the latter bound is also valid for the case of uncorrelated sources; however, the former is stricter.

B. Number of Sources Known

First, we consider the training strategy introduced in Section IV-A. For each testing example the output of the CNN is given by (13). Assuming that we know the number of sources, K, we can pick up the K largest probabilities in (13); the indices corresponding to the grid are the estimated DoAs.

The metric that is employed for the evaluation of the CNN is the empirical RMSE, which is defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{D_{\mathrm{te}}K}\sum\nolimits_{k=1}^{K}\sum_{m=1}^{D_{\mathrm{te}}}\left(\theta_k^{(m)} - \hat{\theta}_k^{(m)}\right)^2}, \qquad (14)$$

where $[\theta_1^{(m)}, \ldots, \theta_K^{(m)}]^T$ are the actual DoAs (ground truth) and $[\hat{\theta}_1^{(m)}, \ldots, \hat{\theta}_K^{(m)}]^T$ are the estimated DoAs at the m-th testing example, whereas D_{te} is the total number of testing examples for each experiment.

1) Direction-of-Arrival Estimation and Errors: In the first set of experiments, we consider three DoA scenarios in the low SNR regime. First, two signals with an angular distance of $\Delta\theta = 4.7^\circ$ at $\mathrm{SNR} = -10$ dB impinge onto the array, while the direction of the first signal varies from -60° to 55° with an increasing step of 1° (narrow grid). The angular distance between the two sources is not contained in the training set, hence, the angular direction of the second signal deviates from those used for the training of the network. We considered that $T = 2000$ snapshots are collected for the SCM estimate. The DoAs predicted by the proposed CNN are depicted in Fig. 3(a) (the solid line corresponds to the actual DoAs), whereas in Fig. 3(h) the corresponding errors are plotted. Additionally, in Fig. 3(b), (d) and (e), we have plotted the MUSIC, $\ell_{2,1}$-SVD and MLP DoA estimates, respectively. The directions estimated by R-MUSIC are depicted in Fig. 3(c) with their errors shown in Fig. 3(i). Finally, ESPRIT's and UnESPRIT's DOA estimates are depicted in Fig. 3(f), (g), and their corresponding errors in Fig. 3(j), (k), respectively.

We observe that, the CNN's errors lie in the interval $[-0.3^\circ, 0.7^\circ]$. MUSIC has some very large errors at the edges of the angular region; $\ell_{2,1}$-SVD has errors with higher variance and MLP fails to consistently estimate two DoAs at each sample index (black circle). R-MUSIC and ESPRIT demonstrate overall good performance, but they appear weaker near the edge of the interval (higher variation near $\{-60^\circ, 60^\circ\}$) with errors up to $\pm 2^\circ$. This weakness is alleviated by UnESPRIT, due to its forward-backward averaging property. It's errors lie in the interval $[-1^\circ, 1^\circ]$. Another interesting fact is the difference in the distribution of the CNN errors in Fig. 3(h), which are more uniformly distributed across the angular region compared to the errors of the classical estimators in Fig. 3(i), (j) and (k). Overall, the empirical RMSE in the DoA estimation is 0.30° for the CNN; 5.01° for MUSIC; 0.35° for R-MUSIC; 1° for $\ell_{2,1}$-SVD; 0.35° for ESPRIT and 0.27° for UnESPRIT. It should be noted that the MLP could not resolve DoAs in 18 out of the 116 examples (circled DoA estimates in Fig. 3(e)); therefore, the estimation error cannot be calculated. The threshold value used for the $\ell_{2,1}$-SVD method in this experiment is $\eta = 550$.

Next, we consider two signals with an angular distance of $\Delta\theta = 2.11^\circ$ at $\mathrm{SNR} = 0$ dB. The direction of the first signal varies from -59.5° to 57.5° with an increasing step of 1° (narrow grid). In the second scenario, both signals impinge onto the array from directions unseen during the training procedure of the CNN. The directions are estimated from the SCM estimate using $T = 200$ snapshots. The CNN's DoA estimates are depicted in Fig. 4(a) and the corresponding errors are plotted in Fig. 4(h). Moreover, in Fig. 4(b) and 4(c), we have plotted the MUSIC and R-MUSIC DoA estimates, respectively. In Fig. 4(d), the $\ell_{2,1}$-SVD DoA estimates are depicted and their corresponding errors are shown in Fig. 4(i). The DoAs estimated by the MLP are shown in Fig. 4(e). The ESPRIT and UnESPRIT DOA estimates are depicted in Fig. 4(f), (g), and their corresponding errors in Fig. 4(j), (k), respectively.

Despite the small angular separation between the angles, we observe that all CNN's errors lie in the interval $[-0.5^\circ, 0.61^\circ]$. On the other hand, MUSIC has several large errors and R-MUSIC has only a few large errors (circled) close to the boundaries of the angular region, i.e., $\{-60^\circ, 60^\circ\}$. The performance of the $\ell_{2,1}$-SVD appears to be more robust and similar to that of the proposed CNN. However, we should note that the tuning of the η threshold (noise bound) is one of the major disadvantages of the $\ell_{2,1}$-SVD method, since in many practical applications the

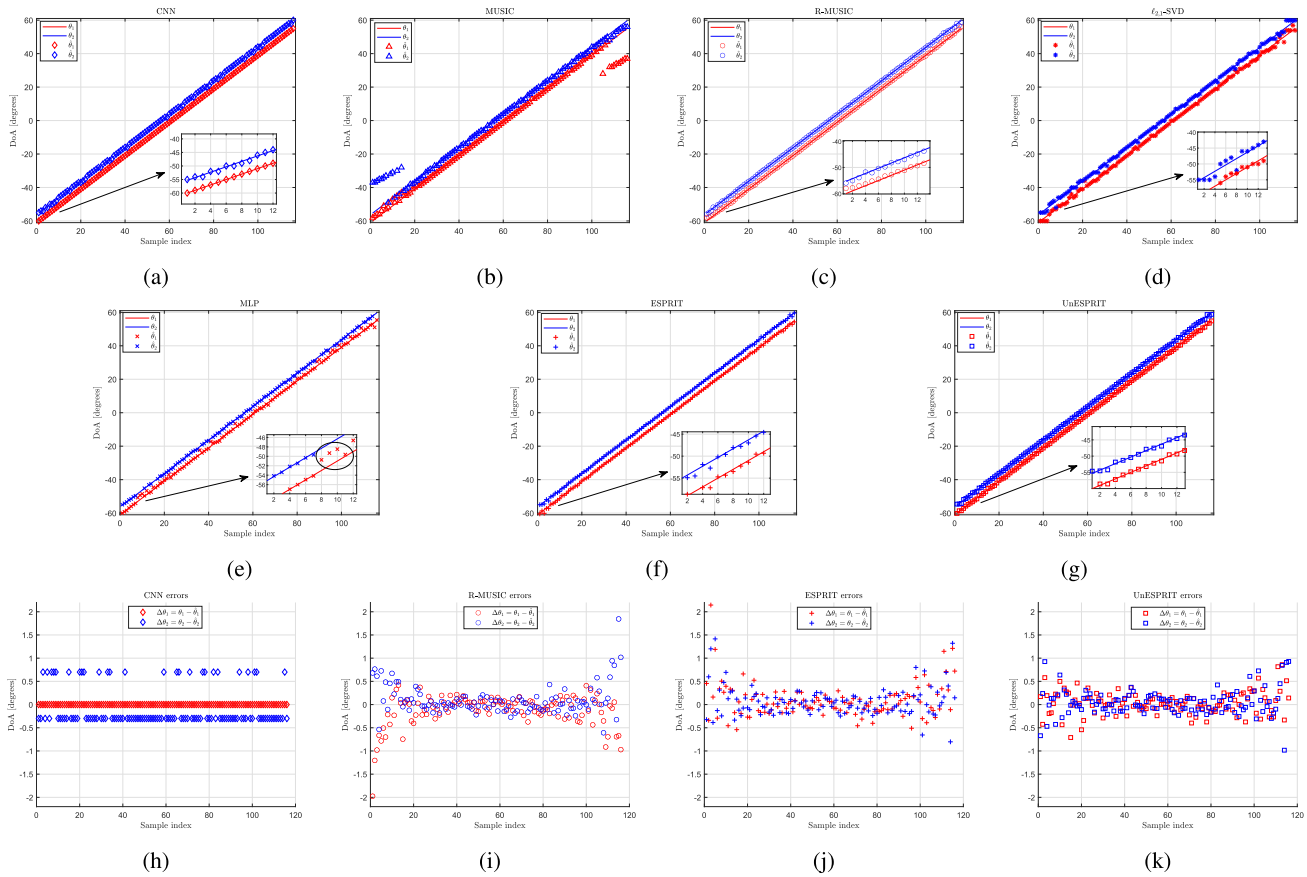


Fig. 3. DoA estimation performance on off-grid angles $\theta_1,\ \theta_2 \in [-60^\circ, 60^\circ]$ at -10 dB SNR using $T = 2000$ snapshots. The DoAs of: (a) CNN (proposed), (b) MUSIC, (c) R-MUSIC, (d) $\ell_{2,1}$-SVD, (e) MLP, (f) ESPRIT and (g) UnESPRIT. The DoA estimation errors of the: (h) CNN, (i) R-MUSIC, (j) ESPRIT and (k) UnESPRIT. The CNN outperforms its competitors in DoA estimation i) attaining the smallest errors and ii) demonstrating a robust performance comparable to that of the Root-MUSIC and the Unitary ESPRIT estimator.

operating SNR level is unknown.[5] The MLP demonstrates poor performance failing to resolve the two closely spaced angles in the current setup 103 out of 118 testing examples (the appearance of large errors is secondary). Although, ESPRIT and UnESPRIT show improvement compared to MUSIC and R-MUSIC, they still have quite large errors near the edge of the interval. The RMSE in the DoA estimation is 0.50° for the CNN; 20.31° for MUSIC; 11.17° for R-MUSIC; 0.54° for $\ell_{2,1}$-SVD; 1.23° for ESPRIT and 0.83° for UnESPRIT. Hence, we conclude that in contrast to the its competitors, the proposed CNN demonstrates a robust and automated (parameter independent) solution in the DoA estimation of two sources in the low and moderate SNR regime. The threshold value used for the $\ell_{2,1}$-SVD method in this experiment is $\eta = 60$.

In the third experiment, we consider the case where the two DoAs in a wider range, i.e., in $[-90^\circ, 90^\circ]$. We follow a similar setup as in the first experiment. The direction θ_1 of the first signal varies from -90° to 85°, whereas $\theta_2 = \theta_1 + 4.7^\circ$; the SNR is -10 dB and $T = 2000$. Since in this case the grid $\mathcal{G}_w$ has 181 points (instead of 121), the proposed CNN needs to be trained again in a similar manner. The same is true for the MLP as well. Finally, the threshold for $\ell_{2,1}$-SVD was set to $\eta = 550$. Fig. 5 shows the DoA estimates for the case where $\theta_1,\ \theta_2 \in [-90^\circ, 90^\circ]$. We observe that the performance of the proposed CNN is remarkable, especially near the edge of the angular region (endfire) where the estimation is challenging. All classical DoA estimators, i.e., MUSIC, R-MUSIC, ESPRIT and UnESPRIT, fail to provide accurate estimates near the edge. The MLP fails to resolve the angles 85 out of 176 testing examples and has also large errors near the edge. Although, the $\ell_{2,1}$-SVD demonstrates robustness to some extent, it still produces a few large errors near the edges. The RMSE of the estimation is 0.96° for the CNN; 44.69° for MUSIC; 32.22° for R-MUSIC; 11.73° for $\ell_{2,1}$-SVD; 30.12° for ESPRIT, and 26.49° for UnESPRIT. The absolute maximum (worst) error of the CNN is 5.7° whereas its competitors experience large errors near the edges of up to 160° (absolute maximum).

2) RMSE Versus the SNR: In this experiment, we evaluate the performance of the proposed CNN in the DoA estimation of two sources in the directions $\theta_1 = 10.11^\circ$ and $\theta_2 = 13.3^\circ$. At each SNR level the RMSE is calculated over $D_{\text{te}} = 1000$ Monte Carlo (MC) runs, whereas the sample covariance is estimated using $T = 1000$ snapshots. The RMSE results are plotted in

[5]There are of course methods that estimate the SNR level, but correspondence to the η threshold is not straightforward.

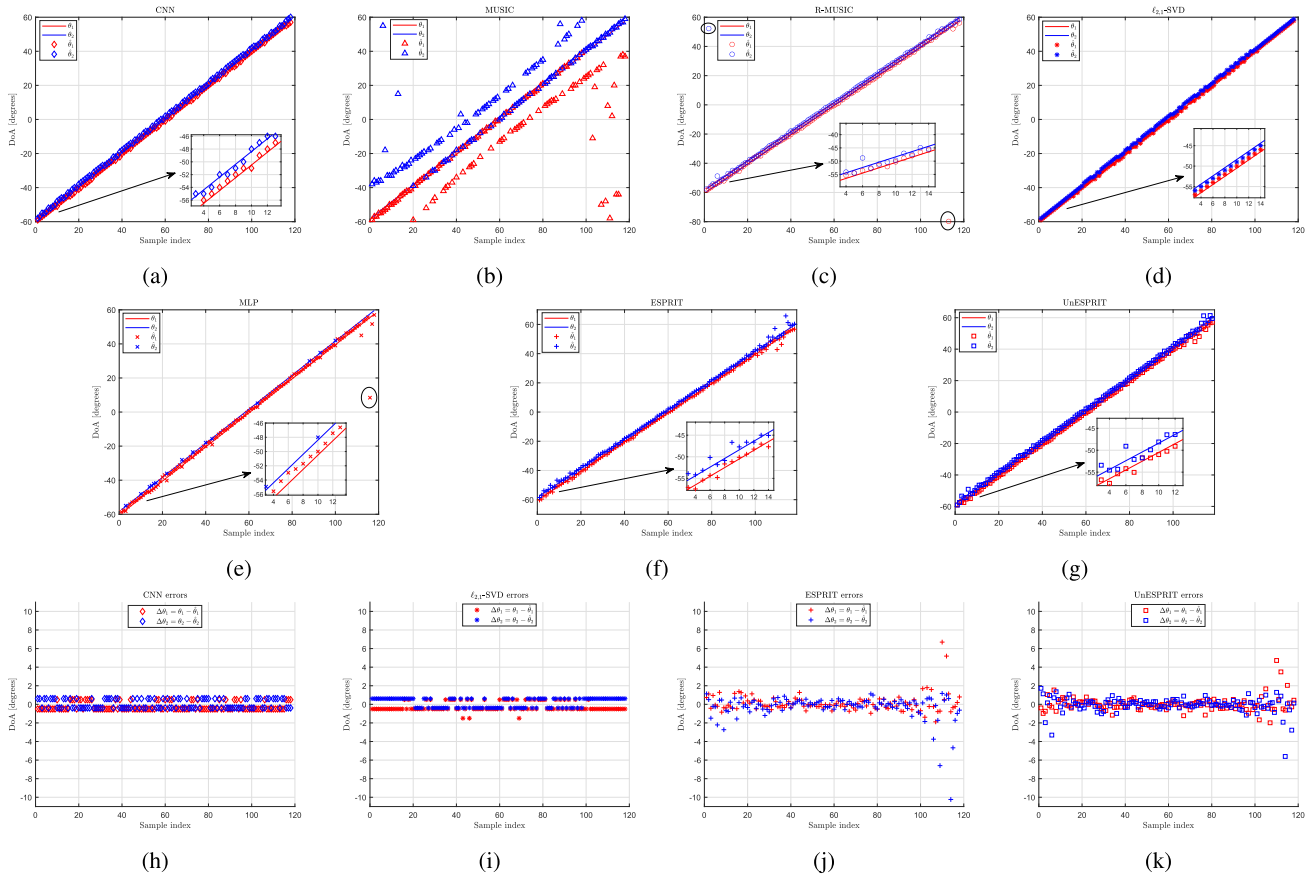


Fig. 4. DoA estimation performance on off-grid angles $\theta_1,\ \theta_2 \in [-60^\circ, 60^\circ]$ at 0 dB SNR using $T = 200$ snapshots. Estimated DoAs: (a) CNN (proposed), (b) MUSIC, (c) R-MUSIC, (d) $\ell_{2,1}$-SVD, (e) MLP, (f) ESPRIT and (g) UnESPRIT. The DoA estimation errors of the: (h) CNN, (i) $\ell_{2,1}$-SVD, (j) ESPRIT and (k) UnESPRIT. The CNN demonstrates robustness in the DoA estimation in low SNR despite the small number of snapshots. Its performance is close to that of the $\ell_{2,1}$-SVD. More importantly, the additional advantage over the latter method is that no threshold tuning is required.

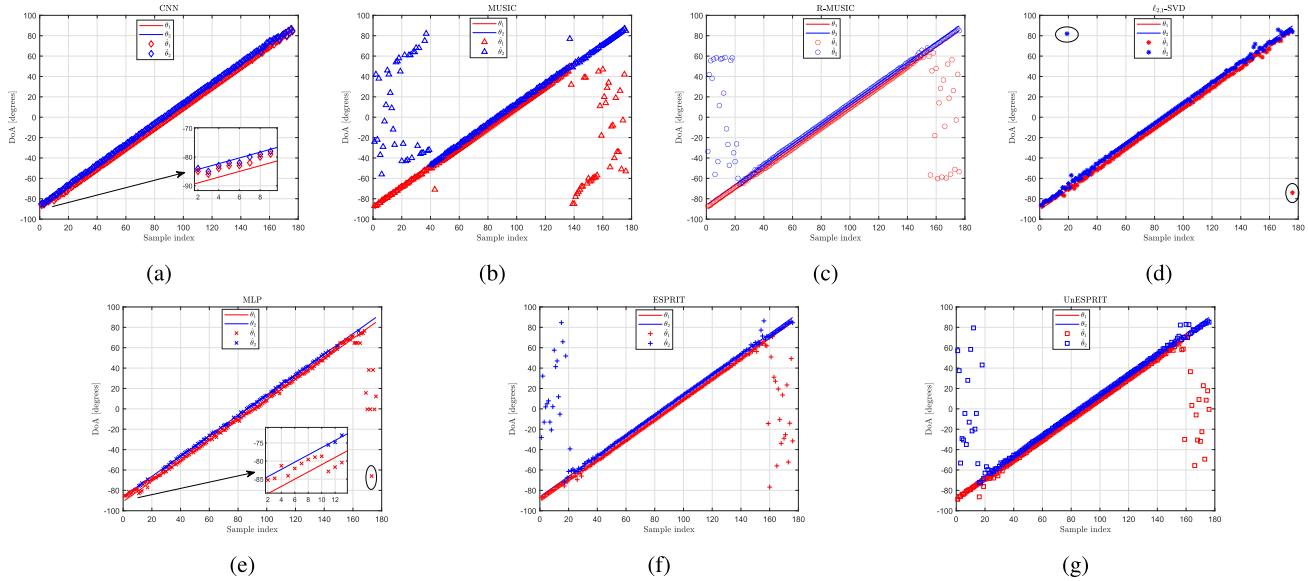


Fig. 5. DoA estimation performance on off-grid angles $\theta_1,\ \theta_2 \in [-90^\circ, 90^\circ]$ at -10 dB SNR using $T = 2000$ snapshots. The DoAs of: (a) CNN (proposed), (b) MUSIC, (c) R-MUSIC, (d) $\ell_{2,1}$-SVD, (e) MLP, (f) ESPRIT and (g) UnESPRIT. The CNN outperforms all its competitors in the DoA estimation of two sources in a wide range of angles demonstrating remarkable resilience to estimation near the edge (endfire).

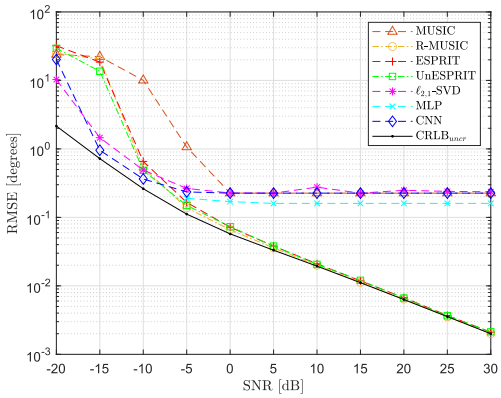


Fig. 6. The RMSE (logarithmic scale) vs the SNR in the DoA estimation of two sources for off-grid angles using $T = 1000$ snapshots. The proposed CNN outperforms the subspace-based methods in the low-SNR regime. In the high-SNR regime the RMSE of all on-grid approaches floors and only the grid-less estimators (ESPRIT, R-MUSIC and UnESPRIT) perform optimally attaining the CRLB.

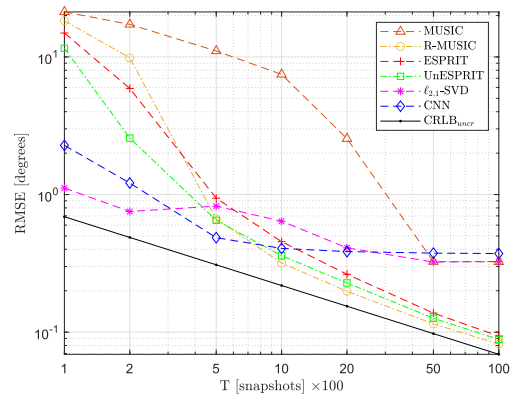


Fig. 7. The RMSE vs the number of snapshots T (both axes in logarithmic scale) in the DoA estimation of two sources for off-grid angles at -10 dB SNR. The proposed CNN demonstrates improved robustness for a small number of snapshots. For a large number of snapshots the performance of grid-based estimators floors, due to the finite grid resolution.

Fig. 6; additionally, the CRLB$_{uncr}$ in [39] is calculated per SNR. In the low-SNR regime, the CNN demonstrates very good performance in terms of the RMSE close to that of the robust $\ell_{2,1}$-SVD. However, in contrast to the later method, no tuning of any sort of parameters is required for the CNN, which is a major advantage in practical applications, where the SNR level is either unknown or it may slightly vary. The results also indicate that in the high-SNR regime the RMSE floors for all on-grid methods, whereas only the grid-less estimators, ESPRIT, R-MUSIC and UnESPRIT attain the CRLB. This performance holds for all grid-based methods and cannot be improved unless a finer grid is utilized. It should be noted that the CNN manages to predict sufficient angle estimates at the high-SNRs despite not being trained in such scenario. We should also not that for the MLP RMSE results below -5 dB SNR are not plotted, since the DoAs could not be (consistently) resolved (single DoA estimate in at least one out of the thousand test examples). The threshold values used for the $\ell_{2,1}$-SVD method in this experiment are $\eta = \{1260, 700, 400, 230, 140, 100, 70, 70, 60, 60, 60\}$ for the corresponding SNR values.

3) RMSE Versus the Number of Snapshots T: In this setup, we attempt to estimate the DoAs of two sources at -10 dB SNR while the number of snapshots, T, varies from 100 to 10000. For each value of T the RMSE is averaged over 1000 MC runs. The direction of the first source is $-13.18°$ and the direction of the second source is $-9.58°$ (off-grid angles). In Fig. 7, the RMSE of the estimation is depicted for each method (both axes are in logarithmic scale). It is observed that for a (relatively) small number of snapshots (up to $T = 500$) the CNN demonstrates a robust behavior compared to all classical DoA estimators (MUSIC, R-MUSIC, ESPRIT, UnESPRIT). As the number of snapshots increases, the finite resolution of the grid limits the performance of the grid-based estimators. Notable is also the fact that the performance of the CNN is similar to that of the $\ell_{2,1}$-SVD method, but with the additional advantage of not depending on the fine-tuning of any parameters that control the estimation. The MLP was not able to resolve

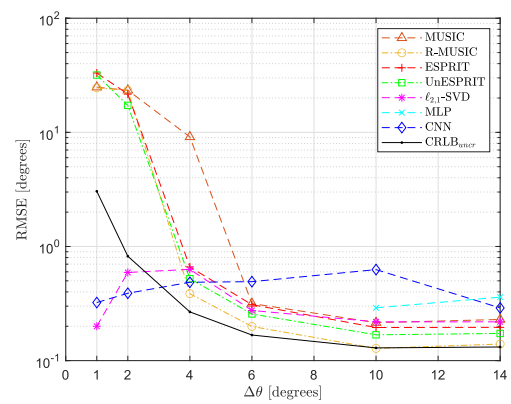


Fig. 8. The RMSE vs the angle separation $\Delta\theta$ in the DoA estimation of two sources for off-grid angles at -10 dB SNR using $T = 500$ snapshots. The small angular separation of the sources does not affect the performance of the CNN and the $\ell_{2,1}$-SVD.

the angles in this setup (therefore not included). The threshold values used for the $\ell_{2,1}$-SVD method in this experiment are $\eta = \{130, 180, 270, 410, 570, 910, 1280\}$ for the corresponding values of T.

4) RMSE Versus the Angle Separation $\Delta\theta$: Fig. 8 includes RMSE results in the DoA estimation of two sources for various angle separations $\Delta\theta \geq 1°$. For each value of $\Delta\theta$ the RMSE is averaged over 1000 MC runs. The DoA of the first source is $\theta_1 = -13.8°$ (off-grid) and the second direction is $\theta_2 = \theta_1 + \Delta\theta$, while SNR$= -10$ dB and $T = 500$. We observe that for closely separated angles ($1° \leq \Delta\theta < 4°$) the CNN and the $\ell_{2,1}$-SVD are able to resolve the angles, as opposed to the rest of the methods, which fail to provide accurate DoA estimates. Moreover, the performance of the CNN and the $\ell_{2,1}$-SVD is almost constant for all angular separations. One can also observe that for small source separation both $\ell_{2,1}$-SVD and the CNN beat the CRLB. One possible explanation is that the estimators are not unbiased. The MLP can only resolve DoAs with angle separation $\Delta\theta > 6°$ (therefore only results for $10°$ and $14°$ are provided). The

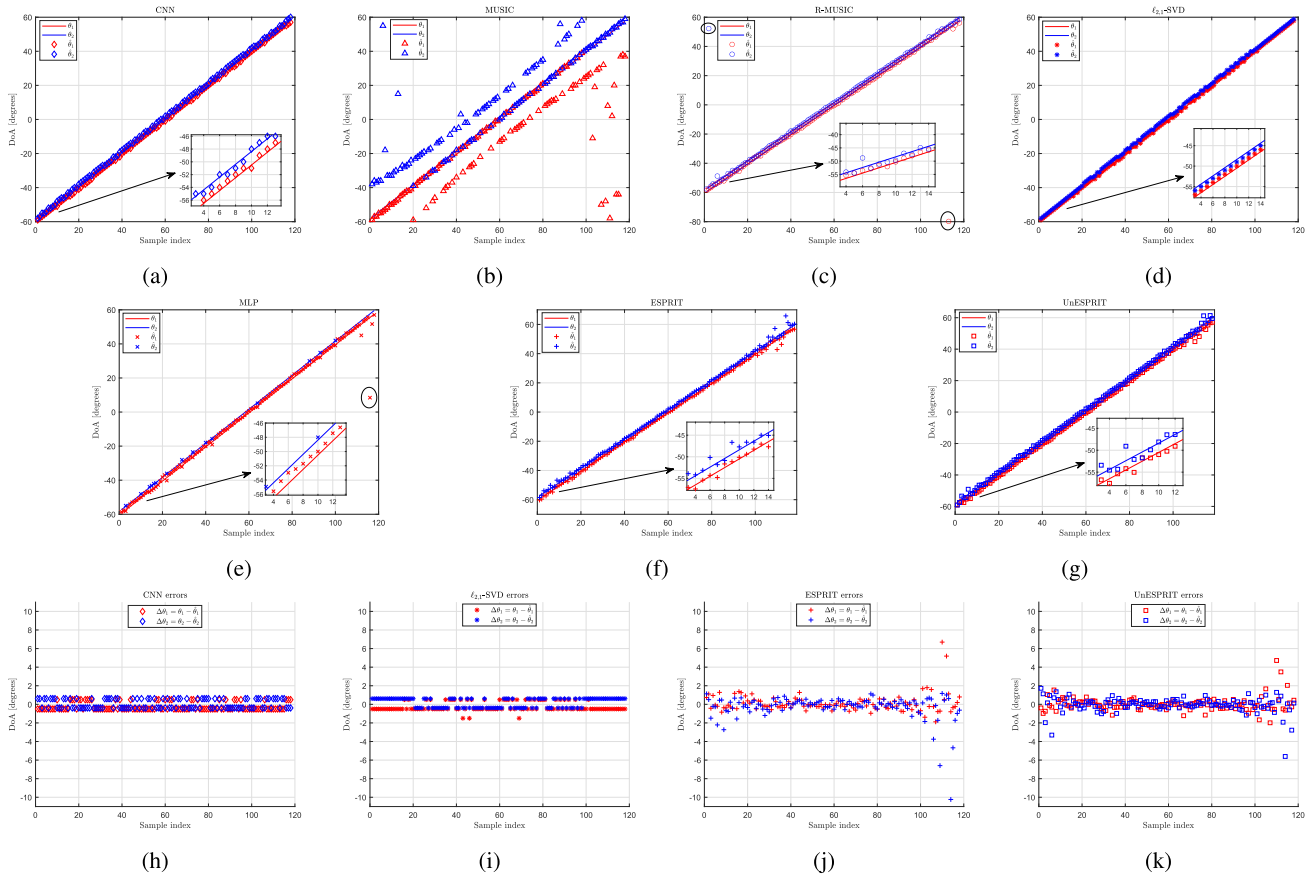


Fig. 9. DoA estimation of two sources under SNR mismatches using $T = 200$ snapshots. The estimated DoAs of: CNN (proposed), (b) MUSIC, (c) R-MUSIC, (d) $\ell_{2,1}$-SVD, (e) MLP, (f) ESPRIT and (g) UnESPRIT. The errors of: (h) CNN, (i) $\ell_{2,1}$-SVD, (j) ESPRIT and (k) UnESPRIT. The CNN attains an RMSE of 0.46°, which is the lowest among its competitors.

threshold value used for the $\ell_{2,1}$-SVD in this experiment is $\eta = 290$.

5) Robustness to SNR Mismatches: In all previous simulations, at each SNR we set $\sigma_1^2 = \sigma_2^2 = 1$ and calculated σ_e^2 according to Eq. (12). However, perfect knowledge of the exact SNR level is not always guaranteed in practice. In this setup, we evaluate the robustness of the proposed method against SNR mismatches. Two scenarios are considered.

The first one closely follows the setup of the second experiment in Section V-B1 at 0 dB SNR (the same angular directions and number of snapshots $T = 200$). The noise variance is $\sigma_e^2 = 1$; however, instead of unit power, we have considered a small perturbation to the sources' power with $\sigma_1^2 = 0.7$ and $\sigma_2^2 = 1.25$. As a result, the actual SNR is now -1.549 dB instead of 0 dB. The DoA estimates for all methods are depicted in Fig. 9. Notable is the performance of the proposed CNN in Fig. 9(a) and $\ell_{2,1}$-SVD in Fig. 9(d). For the latter method the threshold value $\eta = 60$ was optimized for use at 0 dB SNR. We observe a larger dispersion of errors for the $\ell_{2,1}$-SVD, whereas the CNN demonstrates enhanced robustness. The classical estimators, i.e, MUSIC, R-MUSIC, ESPRIT and UnESPRIT, have large errors as we observe in Fig. 9(b), (c), (f), (g), (j) and (k). The MLP could not resolve 71 out of 118 angle pairs. The RMSE of the estimation is: 0.46° for the CNN; 21.46° for MUSIC; 15.79° for R-MUSIC; 0.70° for $\ell_{2,1}$-SVD; 1.49° for ESPRIT, and 0.91° for UnESPRIT.

In the second experiment, we consider two angular directions separated by 4°. The direction of the first source ranges from $-59.43°$ to 55.57° with an increasing step of 1°, whereas the number of snapshots is $T = 1000$. We consider $\sigma_e^2 = 10$ and produce a small perturbation to the sources' power with $\sigma_1^2 = 0.7$ and $\sigma_2^2 = 1.25$, which leads to an actual SNR= -11.549 dB instead of -10 dB. Fig. 10 shows the DoA estimates for all methods. In Fig. 10(a) and (h), we have the DoA estimates and errors of the proposed CNN with an RMSE=0.76°. In Fig. 10(b) and (c), the MUSIC and R-MUSIC DoA estimates are depicted with RMSE equal to 20.26° and 15.85°, respectively. The DoA estimates and errors of the $\ell_{2,1}$-SVD are shown in Fig. 10(d) and (i) with RMSE=1.42° (threshold was set to $\eta = 400$). The MLP DoA estimates can be found in Fig. 10(e); however, once again 41 out of 116 angle pairs were not resolved (hence, no RMSE). Finally, the DoAs of ESPRIT and UnESPRIT are shown in Fig. 10(f), (g) and their errors in Fig. 10(j), (k). Their RMSEs are 1.49° and 1.08°, respectively. From these two experiments, we have evidence that the proposed CNN is resilient to SNR mismatches outperforming all of its competitors.

6) Correlated Signals: So far, we have assumed that the signals of the sources were uncorrelated. In the following

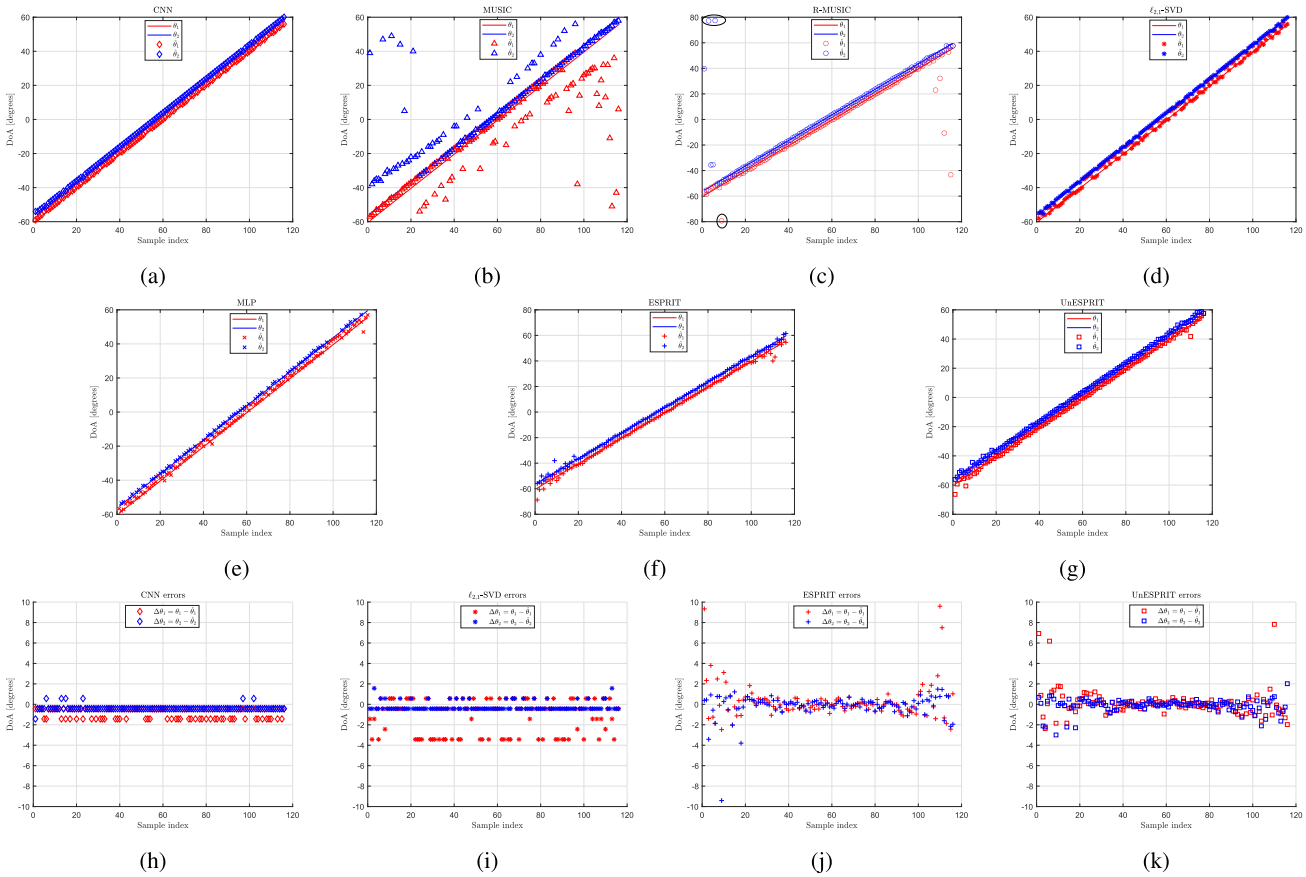


Fig. 10. DoA estimation of two sources under SNR mismatches using $T = 1000$ snapshots. The estimated DoAs of: CNN (proposed), (b) MUSIC, (c) R-MUSIC, (d) $\ell_{2,1}$-SVD, (e) MLP, (f) ESPRIT and (g) UnESPRIT. The errors of: (h) CNN, (i) $\ell_{2,1}$-SVD, (j) ESPRIT and (k) UnESPRIT. The CNN has the smallest errors and attains the lowest RMSE at 0.76°.

experiment, we evaluate the performance of the proposed CNN for the case of correlated signals. In particular, we consider two sources in the directions $\theta_1 = 10^\circ$ and $\theta_2 = 13.8^\circ$ with $\mathbf{R}_s = \left[\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right]$ at -10 dB SNR, whereas $T = 1000$ snapshots are used for the SCM estimate. We let the correlation coefficient ρ vary from 0 (uncorrelated) to 1 (fully correlated signals) and calculate the RMSE for each value of ρ in Fig. 11 (results averaged over $D_{\text{te}} = 1000$ MC runs). Notable is the performance of the CNN (proposed), UnESPRIT and $\ell_{2,1}$-SVD (threshold $\eta = 400$), which are less affected by the increase in the correlation coefficient. However, we should highlight that in low SNR the adopted signal model (unconditional) possesses remarkable insensitivity to the increase of ρ, as mentioned also in [31], something which can be verified by the behavior of the CRLB (constant).

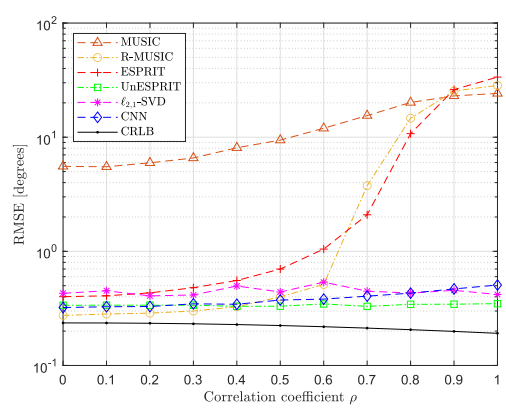


Fig. 11. The RMSE (logarithmic scale) vs the correlation coefficient ρ in the DoA estimation of two sources at -10 dB SNR from $T = 1000$ snapshots. The CNN, UnESPRIT and $\ell_{2,1}$-SVD are not significantly affected by the increase in correlation of the signals.

C. Number of Sources Unknown

In the second part of the section, after training the network according to Section IV-B, we test the CNN without using information about the number of transmitting sources. Thus, we relax the assumption that the number of sources is known a priori and only consider that their maximum number, $K_{\max} = 3$, is known (which is used in the training). After training the model, we can use a confidence level $\bar{p}$ and select the grid points $i = 1, \ldots, 2G+1$ with probability $\hat{p}_i \geq \bar{p}$ in Eq. (13). However, there is no guarantee that the number of source DoAs will now be equal to K; as a matter of fact, the cardinality of the predicted sets of directions may vary from zero to $\hat{K}_{\max}$ (where $\hat{K}_{\max} \lesseqgtr K_{\max}$). Hence, the RMSE metric is no longer

TABLE I
CNN'S DOA ESTIMATION RESULTS WITHOUT KNOWLEDGE ON THE NUMBER OF TRANSMITTING SOURCES

SNR= -10 dB				
Number of sources K	**Confidence level** $\bar{p}$	**Mean** d_{H} $\mu(d_{\mathrm{H}})$	**Max** d_{H} $\max(d_{\mathrm{H}})$	**RMSE** (K known)
1	90%	-	-	0.20°
2	80%	-	-	0.44°
3	60%	1.42°	10.8°	0.61°
SNR= 0 dB				
Number of sources K	**Confidence level** $\bar{p}$	**Mean** d_{H} $\mu(d_{\mathrm{H}})$	**Max** d_{H} $\max(d_{\mathrm{H}})$	**RMSE** (K known)
1	90%	0.20°	0.2°	0.20°
2	80%	0.61°	5.4°	0.44°
3	60%	1.38°	10.8°	0.66°

suitable for the evaluation of the CNN's performance. To this end, we resort to the Hausdorff distance, which for two sets $\mathcal{A}$ and $\mathcal{B}$ is defined as:

$$d_{\mathrm{H}}(\mathcal{A},\mathcal{B}) = \max\{d(\mathcal{A},\mathcal{B}), d(\mathcal{B},\mathcal{A})\}, \tag{15}$$

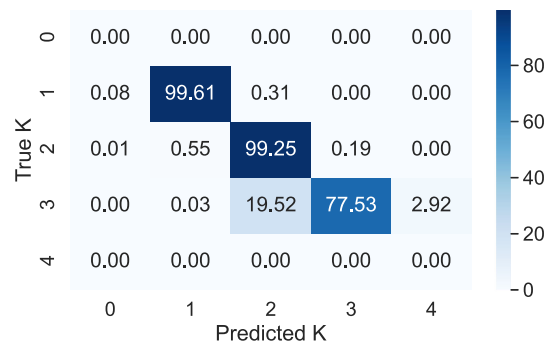
where

$$d(\mathcal{A},\mathcal{B}) = \sup\{d(\alpha,\mathcal{B})|\alpha \in \mathcal{A}\} \tag{16}$$

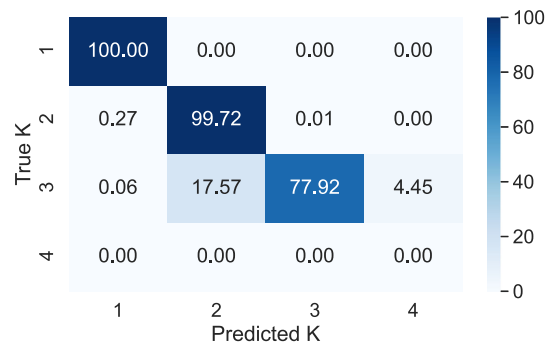
is the directed difference,[6] $d(\alpha,\mathcal{B}) = \inf\{d(\alpha,\beta)|\beta \in \mathcal{B}\}$ and $d(\alpha,\beta) = |\alpha - \beta|$. For the evaluation over the testing set we have used its mean and maximum value, denoted as $\mu(d_{\mathrm{H}})$ and $\max(d_{\mathrm{H}})$, respectively. Simply stated, the Hausdorff distance measures how far two subsets of a metric space are from each other and is particularly important in cases where $\mathcal{A}$ and $\mathcal{B}$ do not share equal cardinality. For example, if $\mathcal{A} = \{-30°, 20°, 23°\}$ and $\mathcal{B} = \{-30.2°, 20.15°, 22.83°\}$ their $\mathrm{RMSE} = 0.18°$ whereas $d_{\mathrm{H}}(\mathcal{A},\mathcal{B}) = 0.2°$ (which is the maximum error corresponding to the first angle). Moreover, if $\mathcal{A} = \{-30°, 21°\}$ the Hausdorff distance is then $d_{\mathrm{H}}(\mathcal{A},\mathcal{B}) = 1.83°$, whereas if $\mathcal{A} = \{-30°, 51°\}$ the Hausdorff distance becomes $d_{\mathrm{H}}(\mathcal{A},\mathcal{B}) = 30.85°$. The latter example shows that large deviations are severely penalized by the metric. Finally, in case one of the two sets is empty the Hausdorff distance becomes infinity.

First, we consider fixed off-grid angles for $K = 1$ up to $K = 3$. The direction of the first signal is 7.8°; the second signal's DoA is at $-2.6°$, and the direction of the third signal is at 2.6° (the angle separation is $\Delta\theta = 5.2°$). We evaluate the performance of the trained CNN in terms of the Hausdorff distance with a) $\mu(d_{\mathrm{H}})$, b) $\max(d_{\mathrm{H}})$, as well as c) on its ability to classify the number of sources. For each $K = 1, 2, 3$ we generated 10000 testing examples using $T = 3000$ and $T = 1000$ snapshots at -10 and 0 dB SNR, respectively. The results of the DoA estimation are reported in Table I at -10 and 0 dB SNR. On the second column we provide the selected confidence level $\bar{p}$; on the third column we have listed the mean Hausdorff distance denoted as $\mu(d_{\mathrm{H}})$; the maximum Hausdorff distance denoted as $\max(d_{\mathrm{H}})$ is listed on the fourth column of the table; finally, on the last column, we have included (for comparison) the RMSE of the estimation assuming that the number of transmitting sources K is known in each case (for comparison only). At -10 dB SNR, for $K = 1, 2$ the mean and max Hausdorff distance cannot be

[6]Notice that in general $d(\mathcal{A},\mathcal{B}) \neq d(\mathcal{B},\mathcal{A})$.



(a) SNR= -10 dB



(b) SNR= 0 dB

Fig. 12. Confusion matrix results (probabilities %) on the number of sources classification using (a) $T = 3000$ and (b) $T = 1000$ snapshots for the SCM estimate. At both SNRs there is a very low error rate for up to $K = 2$, which increases with the number of sources.

evaluated, since in some cases no DoAs were identified, leading to an empty prediction set (with Hausdorff distance equal to infinity). However, these occurrences are quite rare as we will see next. Additionally, in Fig. 12 we have evaluated the network's ability to classify the number of transmitting sources via the confusion matrix. In Fig. 12(a) are the classification results (probabilities %) at -10 dB SNR, whereas in Fig. 12(b) are the results at 0 dB with the confidence levels listed in Table I. Nonzero entries to the right of the main diagonal correspond to Type I or false positive (FP) errors, whereas entries on the left of the main diagonal correspond to Type II or false negative (FN) errors. We observe that, in the majority of the examples, the number of transmitting sources has been correctly identified by the proposed CNN for each $K = 1, 2, 3$. However, we observe that the errors increase with the number of sources, something which is well expected, since the problem is considerably more difficult to solve, especially in the very low SNR.

In the final experiment, we let the directions of the sources vary across the angular region of interest. Specifically, for $K = 1$ we consider 120 test examples with directions from $-59.8°$ to 59.2° and an increasing step of 1°. For $K = 2$, the first signal's direction ranges from $-59.8°$ to 49.2° and the second one's from $-49.8°$ to 59.2° (110 examples with step 1°). For $K = 3$, the first DoA ranges from $-59.8°$ to 39.2°, the second one from $-49.8°$ to 49.2° and the third one from $-39.8°$ to 59.2° (100 examples with step 1°). The testing data are generated at

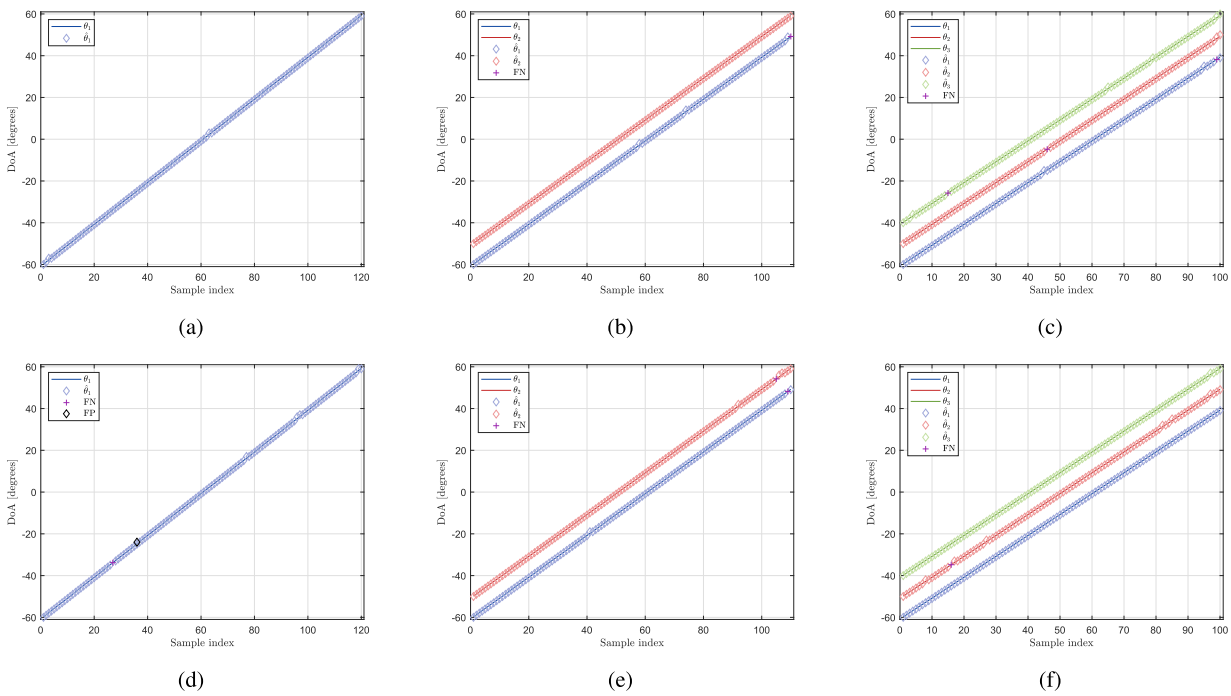


Fig. 13. DoA prediction by the CNN without a priori knowledge on the number K of transmitting sources for varying angles. (a–c) corresponds to SNR$= -10$ dB using $T = 3000$ snapshots. (d–f) corresponds to SNR$= 0$ dB with $T = 1000$ snapshots (two different networks trained at each SNR). In (a), (d) $K = 1$; in (b), (e) $K = 2$ and in (c), (f) $K = 3$.

TABLE II
HAUSDORFF DISTANCE FOR THE DOAS IN THE RANGE $[-60°, 60°]$ WITHOUT KNOWLEDGE ON THE NUMBER OF SOURCES AND CONFIDENCE LEVEL SET TO $\bar{p} = 0.8$.

	SNR$= -10$ dB		SNR$= 0$ dB	
Number of sources K	**Mean** d_{H} $\mu(d_{\mathrm{H}})$	**Max** d_{H} $\max(d_{\mathrm{H}})$	**Mean** d_{H} $\mu(d_{\mathrm{H}})$	**Max** d_{H} $\max(d_{\mathrm{H}})$
1	$0.21°$	$0.80°$	-	-
2	$0.30°$	$9.80°$	$0.40°$	$10.20°$
3	$0.54°$	$10.80°$	$0.33°$	$9.80°$

SNR$= -10$ and SNR$= 0$ dB from $T = 3000$ and $T = 1000$ snapshots, respectively. The results of the DoA estimation without any knowledge of K are depicted in Fig. 13(a–c) at -10 dB SNR and in Fig. 13(d–f) at 0 dB SNR. In Fig. 13(a), we observe that the DoA estimates for $K = 1$ are correct across the entire range of values. In Fig. 13(b), we have one FN (Type II error) occurrence, whereas in Fig. 13(c), we have three FN occurrences. The confidence level for all $K = 1, 2, 3$ was set to $\bar{p} = 0.8$. Despite the low SNR and the fact that the number of transmitting sources is unknown, the proposed CNN manages to estimate the (unknown) directions remarkably well. The results of the same setup at 0 dB SNR are depicted in Figs. 13(d–f). In particular, in Fig. 13(d) where $K = 1$, we have one FP and another FN error. In Fig. 13(e), where $K = 2$, there are two FN errors, whereas in Fig. 13(f) ($K = 3$), there is a single FN error in the estimation of K. Additionally, in Table II, we have calculated the Hausdorff distance of the DoA estimates from the actual directions. Regarding the correctly identified DoAs, they are estimated with a quantization error due the grid mismatch. Of course, the proposed approach with the unknown number of sources can also be applied to the high SNR regime (where we could get sufficient estimates with even smaller number of snapshots) and does not have to be limited to the low SNR region only. However, we would not be able to overcome the grid mismatch error and thus, the results are not expected to improve much compared to those at 0 dB SNR.

VI. CONCLUSION

In this paper, we introduced a deep convolutional neural network (CNN) with 2D filters for DoA prediciton in the low SNR. In particular, we modeled the angle estimation as a multi-label classification task by considering an on-grid approach. The adoption of 2D convolutional layers enables feature extraction from multi-channel input data and information transfer to fully connected layers, leading to robust DoA estimation in low SNR. Two different training strategies are proposed: a) for a fixed and b) for a varying number of sources. The former is suitable in cases where the number of sources is known a priori, which is a typical assumption in the related literature. The proposed solution is compared against state-of-the-art methods and the Cramér-Rao lower bound is also provided as benchmark. The performance of the proposed CNN is evaluated in terms of the RMSE for off-grid angles under various setups: i) fixed and varying directions, ii) across a wide range of SNRs, iii) for various snapshots, iv) various angular separations of the sources and v) in situations of SNR mismatches. The results indicate a) enhanced robustness, b) resilience to the estimation for a wide range of snapshots and c) ability to resolve closely spaced angles in the low SNR.

Additionally, we introduced another training approach for applications in cases where the number of sources is unknown. In particular, we trained the network in the low SNRs and for a varying number of sources, leading to a probabilistic angular estimator. The reported results indicate that the proposed CNN is able to successfully identify the (unknown) number of sources jointly with the DoAs with high probability. Moreover, the predicted angles are sufficiently good estimates of the true DoAs. Possible future research directions include deriving architectures with higher grid resolution (finer grid) for the identification of a large number of targets.

Acknowledgment

The authors would like to thank the anonymous reviewers for their insightful comments and suggestions.

References

[1] R. Chellappa and S. Theodoridis, *Academic Press Library in Signal Processing Volume 3: Array and Statistical Signal Processing*. Elsevier, 2013.

[2] C. Liu and P. P. Vaidyanathan, "Remarks on the spatial smoothing step in coarray music," *IEEE Signal Process. Lett.*, vol. 22, no. 9, pp. 1438–1442, Sep. 2015.

[3] P. Pal and P. P. Vaidyanathan, "Nested arrays: A novel approach to array processing with enhanced degrees of freedom," *IEEE Trans. Signal Process.*, vol. 58, no. 8, pp. 4167–4181, Aug. 2010.

[4] R. Schmidt, "A signal subspace approach to multiple emitter location spectral estimation," Ph.D. dissertation, Stanford Univ., Stanford, CA, USA, 1981.

[5] R. Roy and T. Kailath, "Esprit-estimation of signal parameters via rotational invariance techniques," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 37, no. 7, pp. 984–995, Jul. 1989.

[6] M. Haardt and M. E. Ali-Hackl, "Unitary esprit: How to exploit additional information inherent in the relational invariance structure," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 1994, vol. 4, pp. IV/229–IV/232.

[7] M. Haardt and J. A. Nossek, "Unitary esprit: How to obtain increased estimation accuracy with a reduced computational burden," *IEEE Trans. Signal Process.*, vol. 43, no. 5, pp. 1232–1242, May 1995.

[8] A. Barabell, "Improving the resolution performance of eigenstructure-based direction-finding algorithms," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 1983, vol. 8, pp. 336–339.

[9] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.

[10] Y. C. Eldar and G. Kutyniok, *Compressed Sensing: Theory and Applications*, Cambridge Univ. Press, 2012.

[11] Z. Yang *et al.*, "Sparse methods for direction-of-arrival estimation," *Academic Press Library in Signal Processing*, vol. 7, pp. 509–581. Elsevier, 2018.

[12] J. Chen and X. Huo, "Theoretical results on sparse representations of multiple-measurement vectors," *IEEE Trans. Signal Process.*, vol. 54, no. 12, pp. 4634–4643, Dec. 2006.

[13] Z. Yang, L. Xie, and C. Zhang, "A discretization-free sparse and parametric approach for linear array signal processing," *IEEE Trans. Signal Process.*, vol. 62, no. 19, pp. 4959–4973, Oct. 2014.

[14] Y. Chi, L. L. Scharf, A. Pezeshki, and A. R. Calderbank, "Sensitivity to basis mismatch in compressed sensing," *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2182–2195, May 2011.

[15] Y. C. Eldar and M. Mishali, "Robust recovery of signals from a structured union of subspaces," *IEEE Trans. Inf. Theory*, vol. 55, no. 11, pp. 5302–5316, Nov. 2009.

[16] Z. Yang and L. Xie, "Enhancing sparsity and resolution via reweighted atomic norm minimization," *IEEE Trans. Signal Process.*, vol. 64, no. 4, pp. 995–1006, Feb. 2016.

[17] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436, 2015.

[18] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[19] Y. Kase *et al.*, "DoA estimation of two targets with deep learning," in *Proc. 15th Workshop Positioning, Navigation Commun.*, 2018, pp. 1–5.

[20] H. Huang *et al.*, "Deep learning for super-resolution channel estimation and DOA estimation based massive MIMO system," *IEEE Trans. Veh. Technol.*, vol. 67, no. 9, pp. 8549–8560, Sep. 2018.

[21] Z. Liu, C. Zhang, and P. S. Yu, "Direction-of-arrival estimation based on deep neural networks with robustness to array imperfections," *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 7315–7327, Dec. 2018.

[22] L. Wu, Z. Liu, and Z. Huang, "Deep convolution network for direction of arrival estimation with sparse prior," *IEEE Signal Process. Lett.*, vol. 26, no. 11, pp. 1688–1692, Nov. 2019.

[23] S. Chakrabarty and E. A. P. Habets, "Broadband doa estimation using convolutional neural networks trained with noise signals," in *Proc. IEEE Workshop Appl. Signal Process. Audio Acoust.*, 2017, pp. 136–140.

[24] S. Chakrabarty and E. A. P. Habets, "Multi-speaker DOA estimation using deep convolutional networks trained with noise signals," *IEEE J. Sel. Topics Signal Process.*, vol. 13, no. 1, pp. 8–21, Mar. 2019.

[25] H. Niu, E. Ozanich, and P. Gerstoft, "Ship localization in santa barbara channel using machine learning classifiers," *J. Acoustical Soc. Amer.*, vol. 142, no. 5, pp. EL 455–EL460, 2017.

[26] H. Niu, E. Reeves, and P. Gerstoft, "Source localization in an ocean waveguide using supervised machine learning," *J. Acoustical Soc. Amer.*, vol. 142, no. 3, pp. 1176–1188, 2017.

[27] G. Izacard, B. Bernstein, and C. Fernandez-Granda, "A learning-based framework for line-spectra super-resolution," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2019, pp. 3632–3636.

[28] E. Ozanich, P. Gerstoft, and H. Niu, "A deep network for single-snapshot direction of arrival estimation," in *Proc. IEEE 29th Int. Workshop Mach. Learn. Signal Process.*, 2019, pp. 1–6.

[29] E. Ozanich, P. Gerstoft, and H. Niu, "A feedforward neural network for direction-of-arrival estimation," *J. Acoustical Soc. Amer.*, vol. 147, no. 3, pp. 2035–2048, 2020.

[30] M. J. Bianco *et al.*, "Machine learning in acoustics: Theory and applications," *J. Acoustical Soc. Amer.*, vol. 146, no. 5, pp. 3590–3628, 2019.

[31] P. Stoica and A. Nehorai, "Performance study of conditional and unconditional direction-of-arrival estimation," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1783–1795, Oct. 1990.

[32] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, 1989.

[33] Y. Lecun *et al.*, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[34] J. Gu *et al.*, "Recent advances in convolutional neural networks," *Pattern Recognit.*, vol. 77, pp. 354–377, 2018.

[35] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[36] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2014.

[37] M. Wang and A. Nehorai, "MUSIC Coarrays, and Cramér-Rao the Bound," *IEEE Trans. Signal Process.*, vol. 65, no. 4, pp. 933–946, Feb. 2017.

[38] H. L. Van Trees, *Optimum Array Processing: Part IV of Detection, Estimation, and Modulation Theory*. Wiley, 2004.

[39] M. Jansson, B. Goransson, and B. Ottersten, "A subspace method for direction of arrival estimation of uncorrelated emitter signals," *IEEE Trans. Signal Process.*, vol. 47, no. 4, pp. 945–956, Apr. 1999.

Georgios K. Papageorgiou (Member, IEEE) received the M.Sc. degree in applied mathematics and the Ph.D. degree in informatics and telecommunications from the National and Kapodistrian University of Athens, Greece, in 2012 and 2016, respectively. He is currently a Research Associate with Heriot-Watt University of Edinburgh, U.K. He is a Member of the Research Institute of Sensors, Signals and Systems, one of five Research Institutes forming the research infrastructure of the School of Engineering & Physical Sciences. During 2016–2018, he was a Researcher with Broadband Wireless and Sensor Networks Laboratory, Athens Information Technology, Greece. His main field of expertise is machine learning, signal processing and optimization, and their application to wireless systems and array processing. He has worked on several European funded projects (FP7, H2020), and industrial research projects and has actively participated in over-the-air demos, one of which was the recipient of the Best Booth Award at the EuCNC 2016 conference in Athens.

Mathini Sellathurai (Senior Member, IEEE) is currently a Professor of signal processing and wireless communications and the Dean of science and engineering with Heriot-Watt University, Edinburgh, U.K. She has been active in signal processing research for the past 20 years and has a strong international track record in multiple-input, multiple-output (MIMO) signal processing with applications in radar, and wireless communications. She held visiting positions with Bell-Laboratories, Holmdel, NJ, USA, and with The Canadian Communications Research Centre, Ottawa, ON, Canada. She has authored or coauthored more than 200 peer reviewed papers in leading international journals and IEEE conferences, given invited talks and has written several book chapters and a research monograph as a lead author. Her current research interests include full-duplex systems, passive radar topography, localisation, massive-MIMO, non-orthogonal multiple access, waveform designs, caching technologies, assisted care technologies, IoT, hearing-aids, optimal coded-modulation designs using auto-encoders, channel prediction, and mm-wave imaging and communications. She was the recipient of the IEEE Communication Society Fred W. Ellersick Best Paper Award in 2005, the Industry Canada Public Service Awards for contributions to Science and Technology in 2005, and Awards for contributions to Technology Transfers to Industry in 2004. She was the recipient of the Natural Sciences and Engineering Research Council of Canada (NSERC) Doctoral Award for her Ph.D. dissertation in 2002. She was the Editor of the IEEE TRANSACTIONS ON SIGNAL PROCESSING from 2009 to 2014, and from 2015 to 2018, the General Co-Chair of IEEE SPAWC2016 in Edinburgh, and a Member of the IEEE SPCOM Technical Strategy Committee from 2014 to 2019.

Yonina C. Eldar (Fellow, IEEE) received the B.Sc. degree in physics and the B.Sc. degree in electrical engineering from Tel-Aviv University, Tel-Aviv, Israel, in 1995 and 1996, respectively, and the Ph.D. degree in electrical engineering and computer science from the Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, in 2002.

She is currently a Professor with the Department of Mathematics and Computer Science, Weizmann Institute of Science, Rehovot, Israel. She was previously a Professor with the Department of Electrical Engineering, Technion, where she held the Edwards Chair of engineering. She is also a Visiting Professor with MIT, a Visiting Scientist with Broad Institute, and an Adjunct Professor with Duke University, Durham, NC, USA, and was a Visiting Professor with Stanford. She is the author of the book *Sampling Theory: Beyond Bandlimited Systems* and the coauthor of four other books published by Cambridge University Press. Her research interests include statistical signal processing, sampling theory and compressed sensing, learning and optimization methods, and their applications to biology, medical imaging, and optics.

She is a Member of the Israel Academy of Sciences and Humanities (elected 2017) and a EURASIP Fellow. She was the recipient of many awards for excellence in research and teaching, including the IEEE Signal Processing Society Technical Achievement Award (2013), the IEEE/AESS Fred Nathanson Memorial Radar Award (2014), and the IEEE Kiyo Tomiyasu Award (2016). She was a Horev Fellow of the Leaders in Science and Technology program at the Technion and an Alon Fellow. She was the recipient of the Michael Bruno Memorial Award from the Rothschild Foundation, the Weizmann Prize for Exact Sciences, the Wolf Foundation Krill Prize for Excellence in Scientific Research, the Henry Taub Prize for Excellence in Research (twice), the Hershel Rich Innovation Award (three times), the Award for Women with Distinguished Contributions, the Andre and Bella Meyer Lectureship, the Career Development Chair at the Technion, the Muriel & David Jacknow Award for Excellence in Teaching, and the Technion's Award for Excellence in Teaching (two times). She was the recipient of several best paper awards and best demo awards together with her research students and colleagues, including the SIAM Outstanding Paper Prize, the UFFC Outstanding Paper Award, the Signal Processing Society Best Paper Award, and the IET Circuits, Devices and Systems Premium Award, was selected as one of the 50 most influential women in Israel and in Asia, and is a highly cited Researcher.

She was a Member of the Young Israel Academy of Science and Humanities and the Israel Committee for Higher Education. She is the Editor-in-Chief of the *Foundations and Trends in Signal Processing*, a Member of the IEEE Sensor Array and Multichannel Technical Committee and is on several other IEEE committees. She was a Signal Processing Society Distinguished Lecturer, Member of the IEEE Signal Processing Theory and Methods and Bio Imaging Signal Processing technical committees, and was an Associate Editor for the IEEE TRANSACTIONS ON SIGNAL PROCESSING, the *EURASIP Journal of Signal Processing*, the *SIAM Journal on Matrix Analysis and Applications*, and the *SIAM Journal on Imaging Sciences*. She was Co-Chair and Technical Co-Chair of several international conferences and workshops.